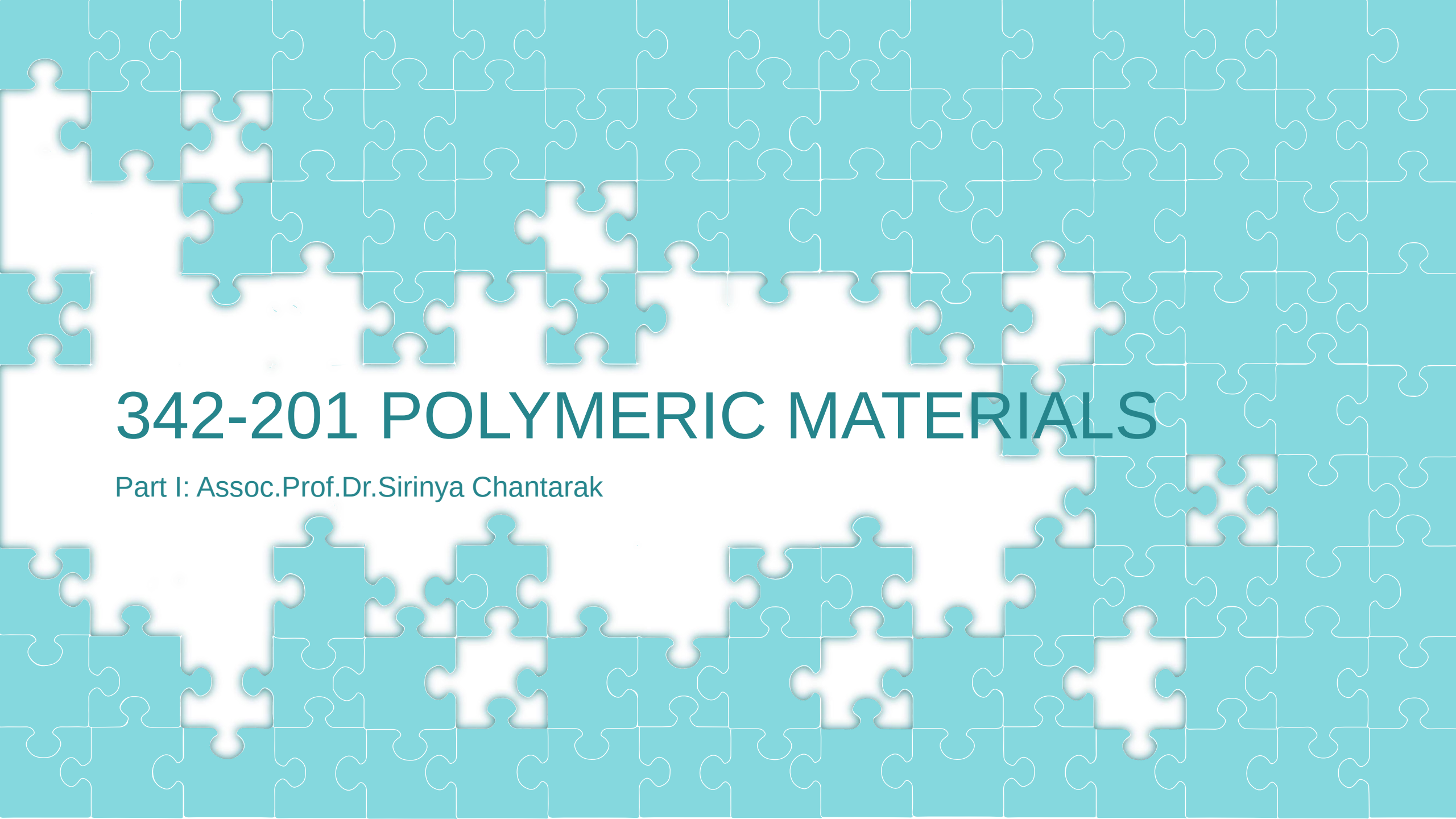


The background of the slide is a repeating pattern of teal-colored puzzle pieces. Some pieces are missing, creating a white, irregular shape in the center of the slide that serves as a backdrop for the text.

# 342-201 POLYMERIC MATERIALS

Part I: Assoc.Prof.Dr.Sirinya Chantarak

# Our Team



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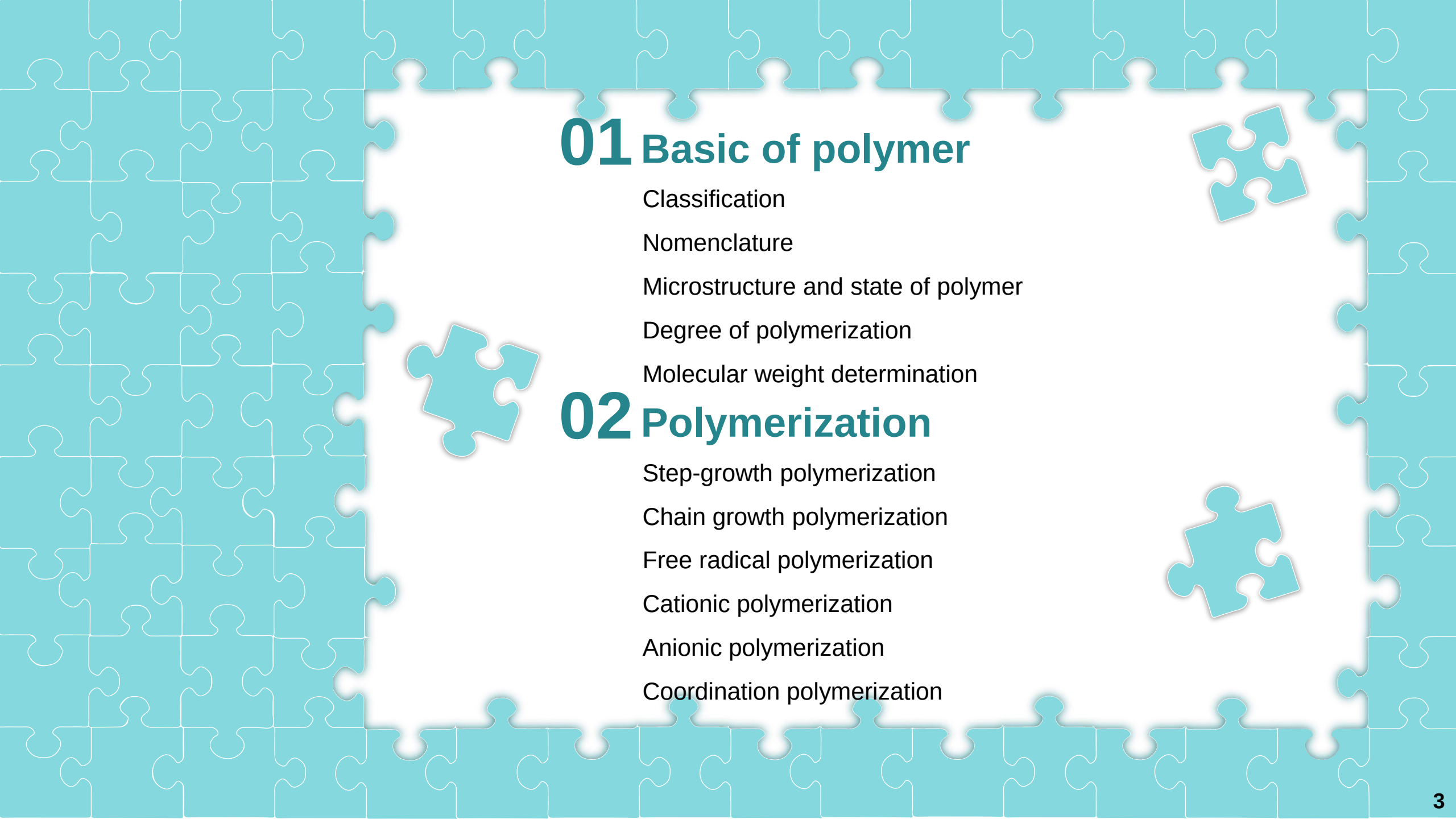
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# 01 Basic of polymer

Classification

Nomenclature

Microstructure and state of polymer

Degree of polymerization

Molecular weight determination

# 02 Polymerization

Step-growth polymerization

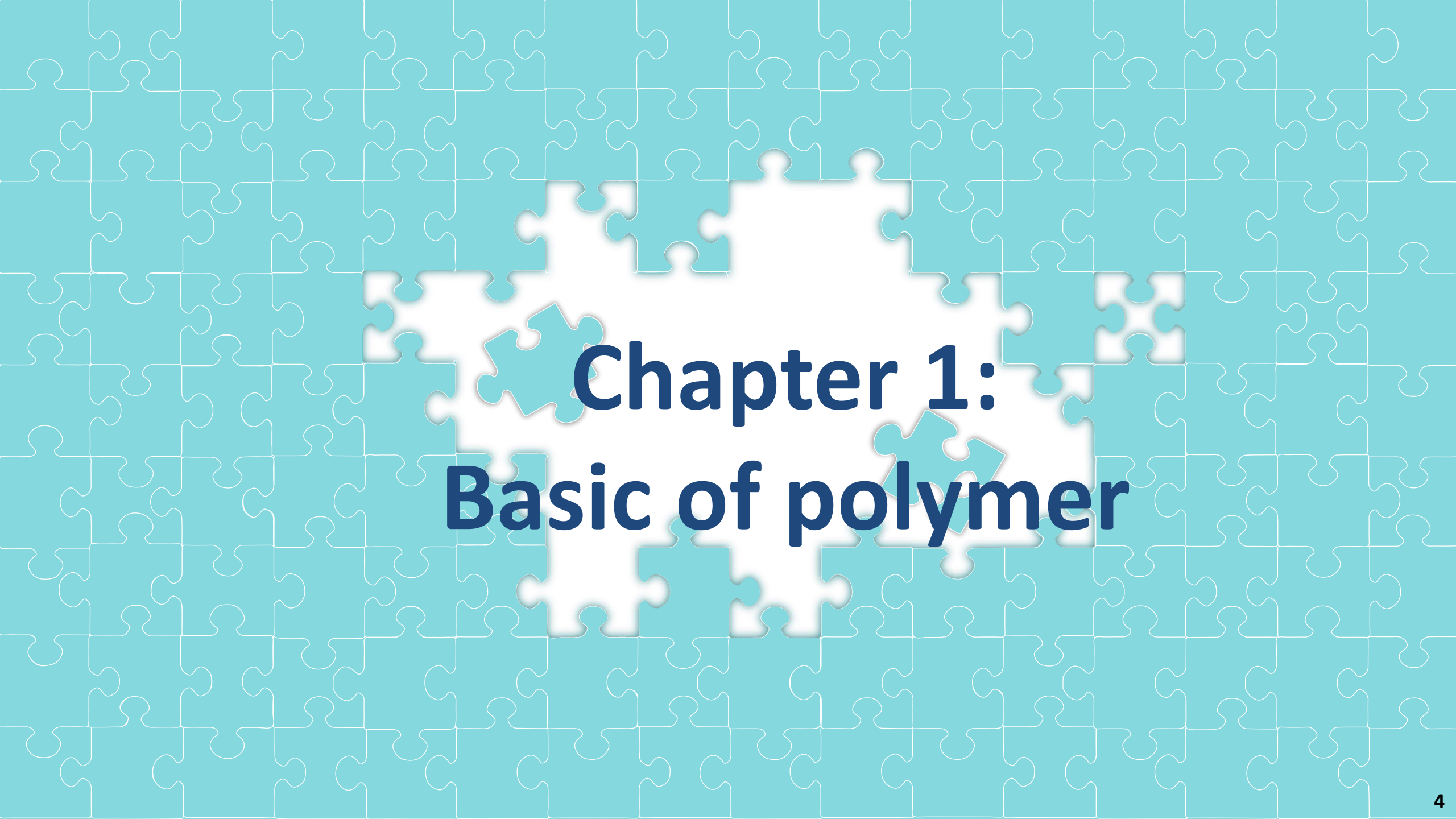
Chain growth polymerization

Free radical polymerization

Cationic polymerization

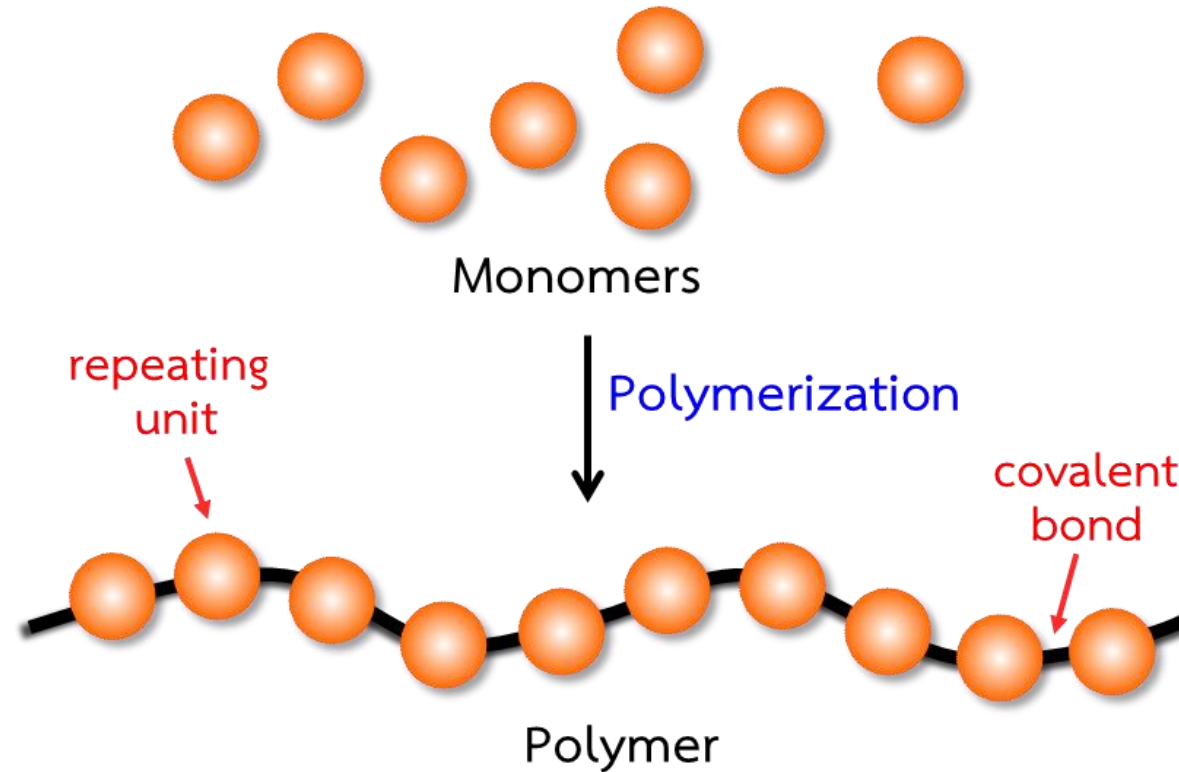
Anionic polymerization

Coordination polymerization



# **Chapter 1:**

# **Basic of polymer**



**poly**  
many

**mer**  
unit

A polymer is a large molecule built up by the repetition of small, simple chemical units held together by covalent bonds.

# Classification of polymer

## Natural

**Organic:** wool, cotton, wood, silk, starch, latex rubber

**Inorganic:** sand, clay



## Synthetic

**Thermoplastics**

**Thermosetting plastics**

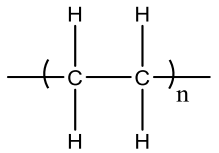


# Thermoplastics

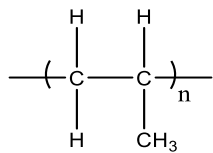
Linear/slightly branched chains molecules capable of repeated softening on heating and hardening on cooling



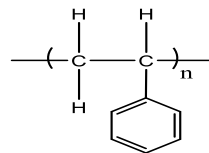
**Polyethylene  
(PE)**



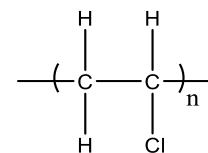
**Polypropylene  
(PP)**



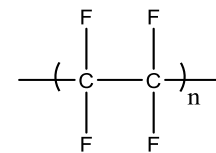
**Polystyrene  
(PS)**



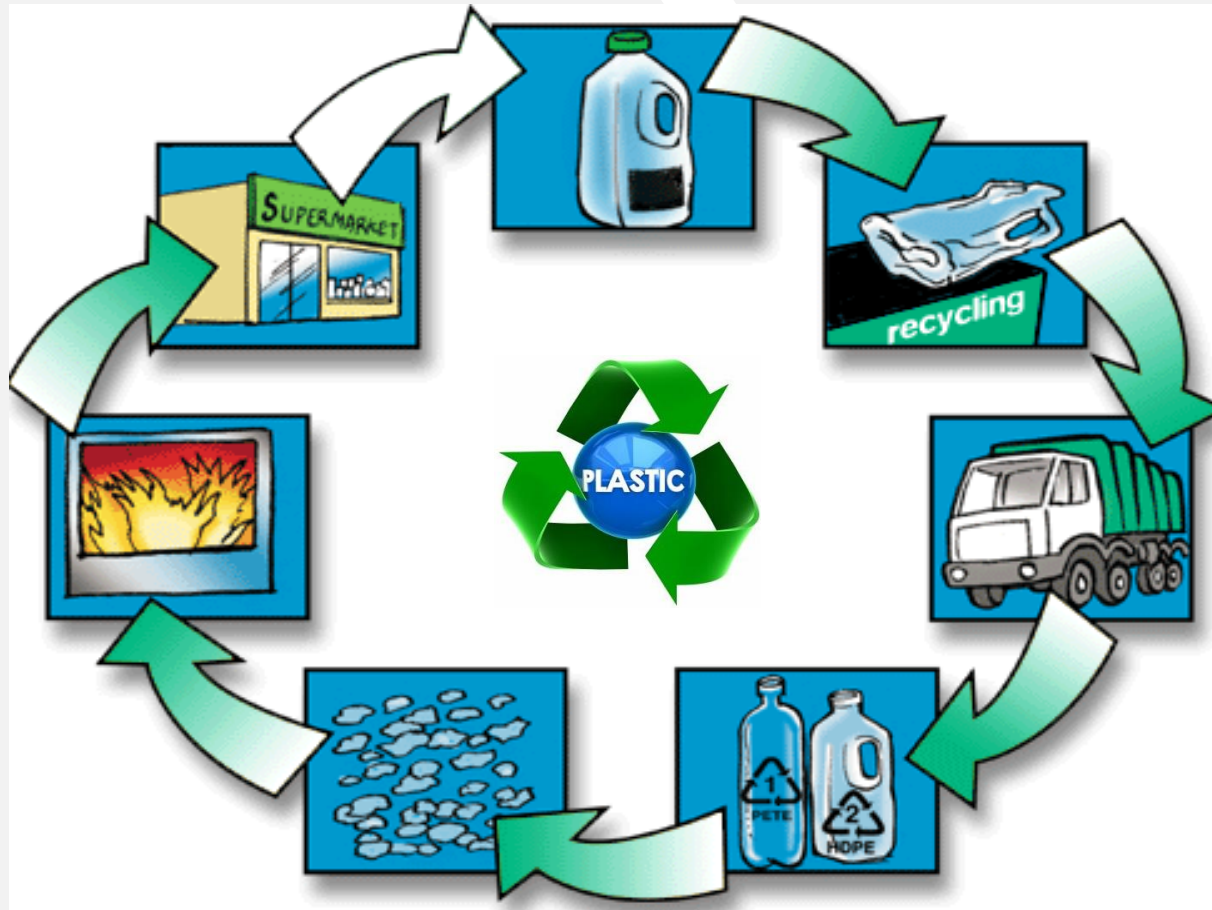
**Poly(vinyl chloride)  
(PVC)**



**Polytetrafluoroethylene  
(PTFE)**



# Recycle process



**PETE**

poly(ethylene terephthalate)



**HDPE**

high-density polyethylene



**PVC**

poly(vinyl chloride)



**LDPE**

low-density polyethylene



**PP**

polypropylene



**PS**

polystyrene



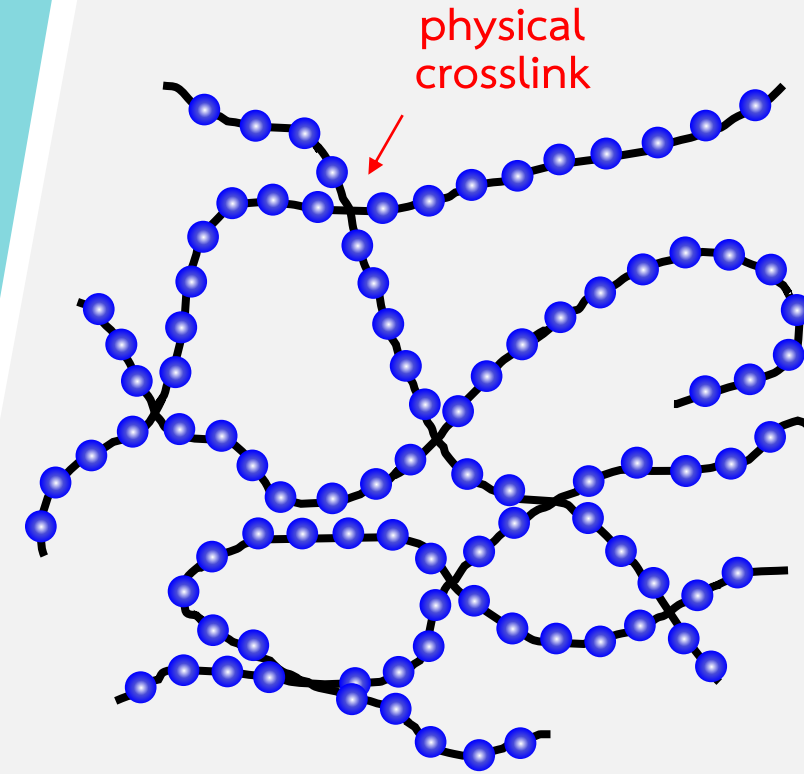
**Others**



bakelite, urea formaldehyde resin

## Elastomers

*flexible polymers contain  
rubbery networks*



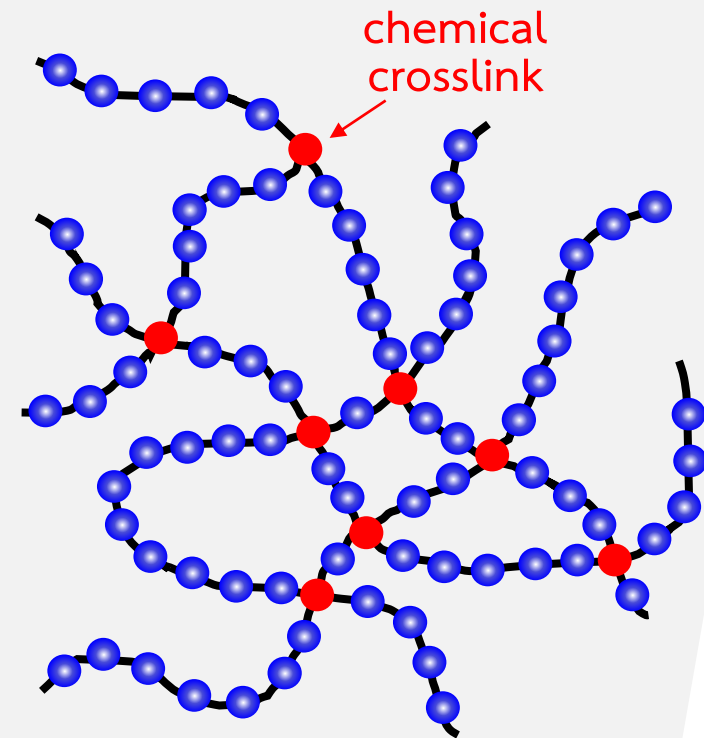
*cis-1,4-polyisoprene*

*styrene  
butadiene  
rubber (SBS)*



## Thermosetting Plastics

*cross-linked or heavily  
branched molecules, which  
on heating undergo  
extensive cross-linkages  
and become infusible*



# Nomenclature: Common name

Add 'poly' as a prefix to the monomer name.

**ethylene**

polyethylene

**styrene**

polystyrene



For complex monomers, parentheses are also added.

**vinyl chloride**

poly(vinyl chloride)

**methyl methacrylate**

poly(methyl methacrylate)

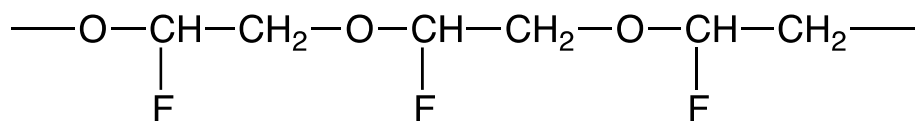
# Common names & applications

| Name(s)                                                       | Formula                      | Monomer                                       | Properties                                                                    | Uses                                        |
|---------------------------------------------------------------|------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------|---------------------------------------------|
| <b>Polyethylene</b><br>low density (LDPE)                     | $-(CH_2-CH_2)_n-$            | ethylene<br>$CH_2=CH_2$                       | soft, waxy solid                                                              | film wrap, plastic bags                     |
| <b>Polyethylene</b><br>high density (HDPE)                    | $-(CH_2-CH_2)_n-$            | ethylene<br>$CH_2=CH_2$                       | rigid, translucent solid                                                      | electrical insulation<br>bottles, toys      |
| <b>Polypropylene</b><br>(PP) different grades                 | $-[CH_2-CH(CH_3)]_n-$        | propylene<br>$CH_2=CHCH_3$                    | <u>atactic</u> : soft, elastic solid<br><u>isotactic</u> : hard, strong solid | similar to LDPE<br>carpet, upholstery       |
| <b>Poly(vinyl chloride)</b><br>(PVC)                          | $-(CH_2-CHCl)_n-$            | vinyl chloride<br>$CH_2=CHCl$                 | strong rigid solid                                                            | pipes, siding, flooring                     |
| <b>Poly(vinylidene chloride)</b><br>(Saran A)                 | $-(CH_2-CCl_2)_n-$           | vinylidene chloride<br>$CH_2=CCl_2$           | dense, high-melting solid                                                     | seat covers, films                          |
| <b>Polystyrene</b><br>(PS)                                    | $-[CH_2-CH(C_6H_5)]_n-$      | styrene<br>$CH_2=CHC_6H_5$                    | hard, rigid, clear solid<br>soluble in organic solvents                       | toys, cabinets<br>packaging (foamed)        |
| <b>Polyacrylonitrile</b><br>(PAN, Orlon, Acrilan)             | $-(CH_2-CHCN)_n-$            | acrylonitrile<br>$CH_2=CHCN$                  | high-melting solid<br>soluble in organic solvents                             | rugs, blankets<br>clothing                  |
| <b>Polytetrafluoroethylene</b><br>(PTFE, Teflon)              | $-(CF_2-CF_2)_n-$            | tetrafluoroethylene<br>$CF_2=CF_2$            | resistant, smooth solid                                                       | non-stick surfaces<br>electrical insulation |
| <b>Poly(methyl methacrylate)</b><br>(PMMA, Lucite, Plexiglas) | $-[CH_2-C(CH_3)CO_2CH_3]_n-$ | methyl methacrylate<br>$CH_2=C(CH_3)CO_2CH_3$ | hard, transparent solid                                                       | lighting covers, signs<br>skylights         |
| <b>Poly(vinyl acetate)</b><br>(PVAc)                          | $-(CH_2-CHOCOCH_3)_n-$       | vinyl acetate<br>$CH_2=CHOCOCH_3$             | soft, sticky solid                                                            | latex paints, adhesives                     |
| <b>cis-Polyisoprene</b><br>natural rubber                     | $-[CH_2-CH=C(CH_3)-CH_2]_n-$ | isoprene<br>$CH_2=CH-C(CH_3)=CH_2$            | soft, sticky solid                                                            | requires vulcanization<br>for practical use |
| <b>Polychloroprene</b> (cis + trans)<br>(Neoprene)            | $-[CH_2-CH=CCl-CH_2]_n-$     | chloroprene<br>$CH_2=CH-CCl=CH_2$             | tough, rubbery solid                                                          | synthetic rubber<br>oil resistant           |

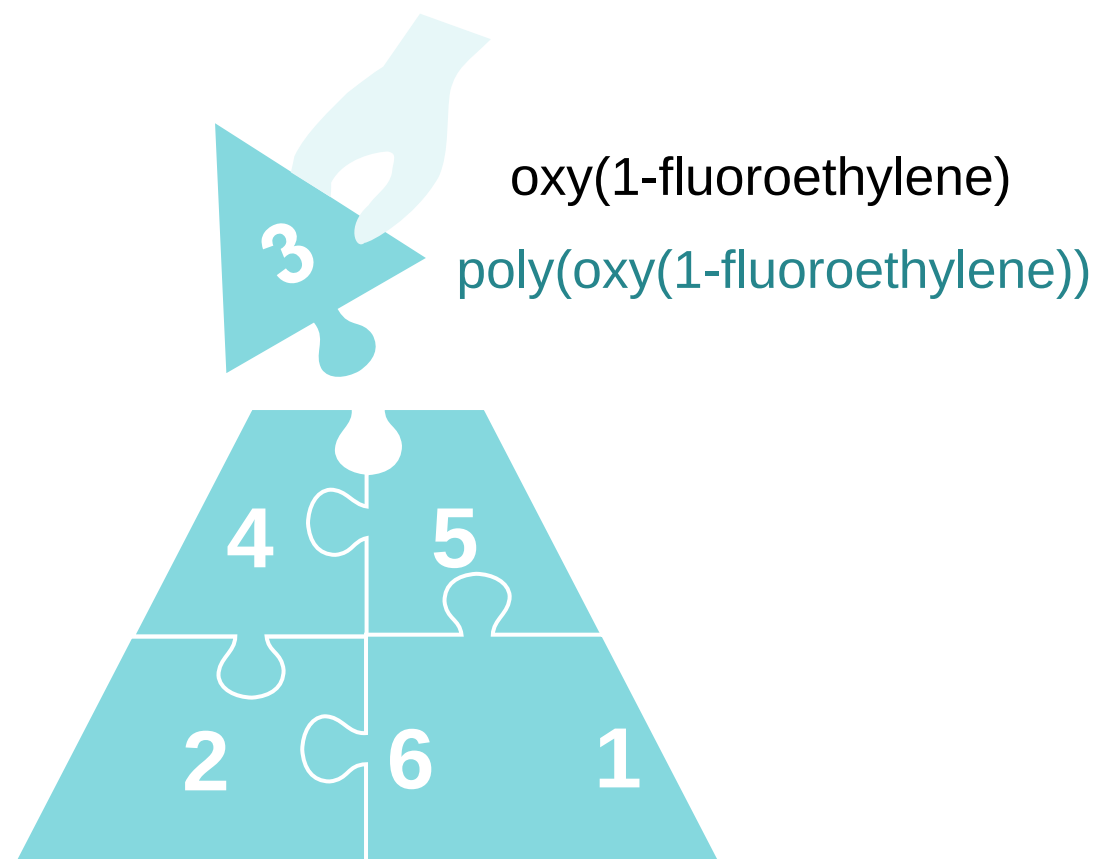
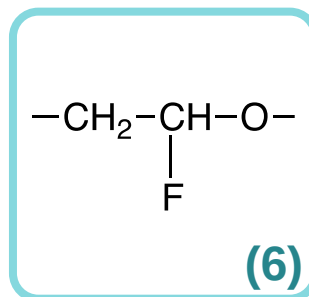
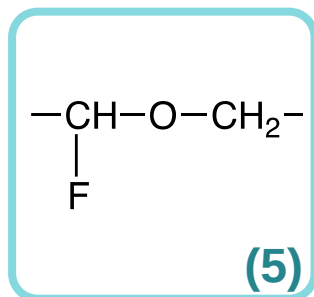
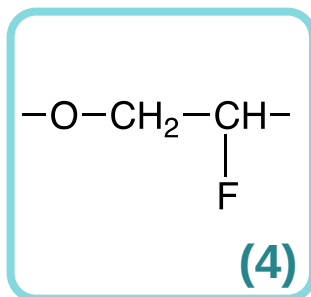
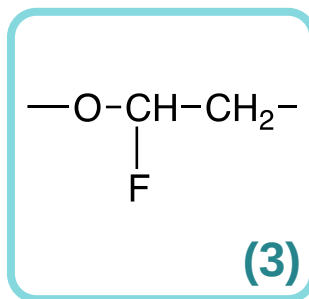
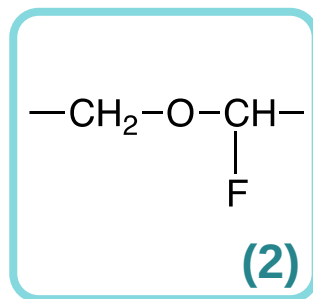
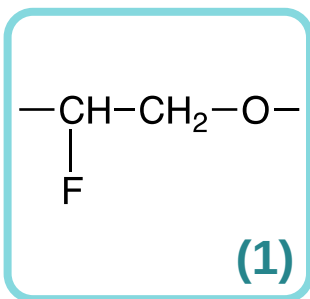


# Nomenclature: IUPAC name

Select a constitutional repeat unit, CRU and orient CRU subunit correctly, following seniority rule. (O, S, Se, Te, N, P, As, Sb, Ni, Si, Ge, Sn, Pb, B, Hg >> C)



6 possible CRUs



# Polymer microstructures

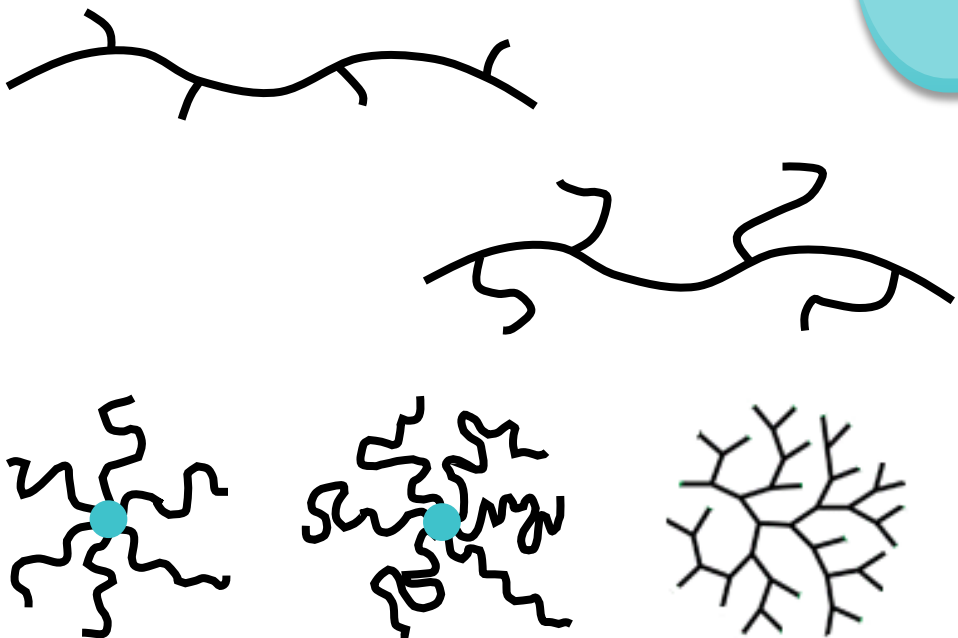
## Branched polymer

short-chained branched polymer

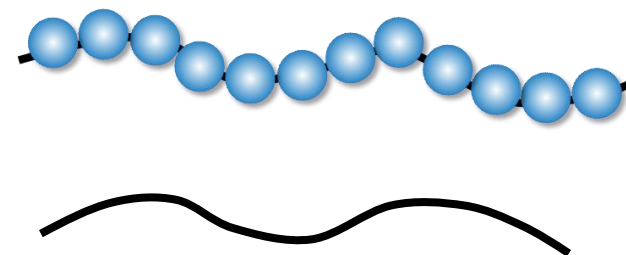
long-chain branched polymer

star polymer

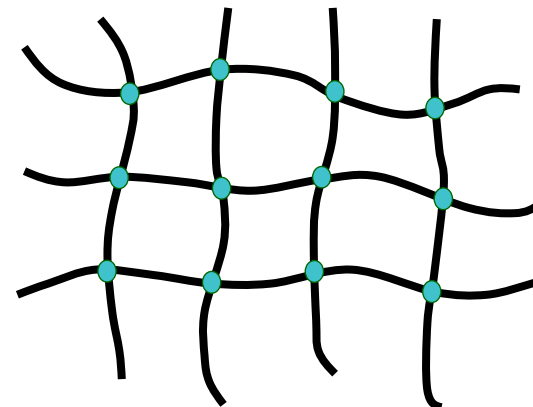
hyper-branched polymer



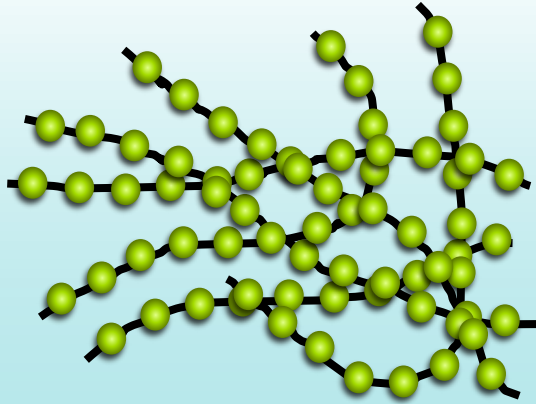
## Linear polymer



## Crosslinked polymer



# State of polymer



## Amorphous polymer

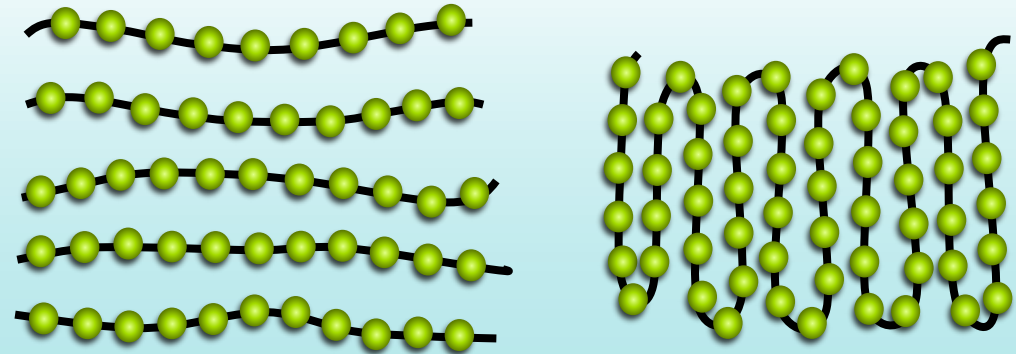
The polymer chains are disordered.

**Glassy**

**Rubbery**

$T_g$

(glass transition temperature)



## Crystalline polymer

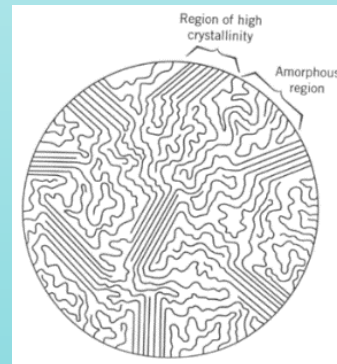
The polymer chains are aligned up.

**Hard**

**Melt**

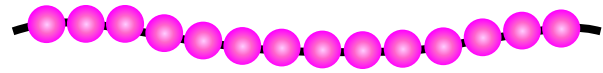
$T_m$

(melting temperature)

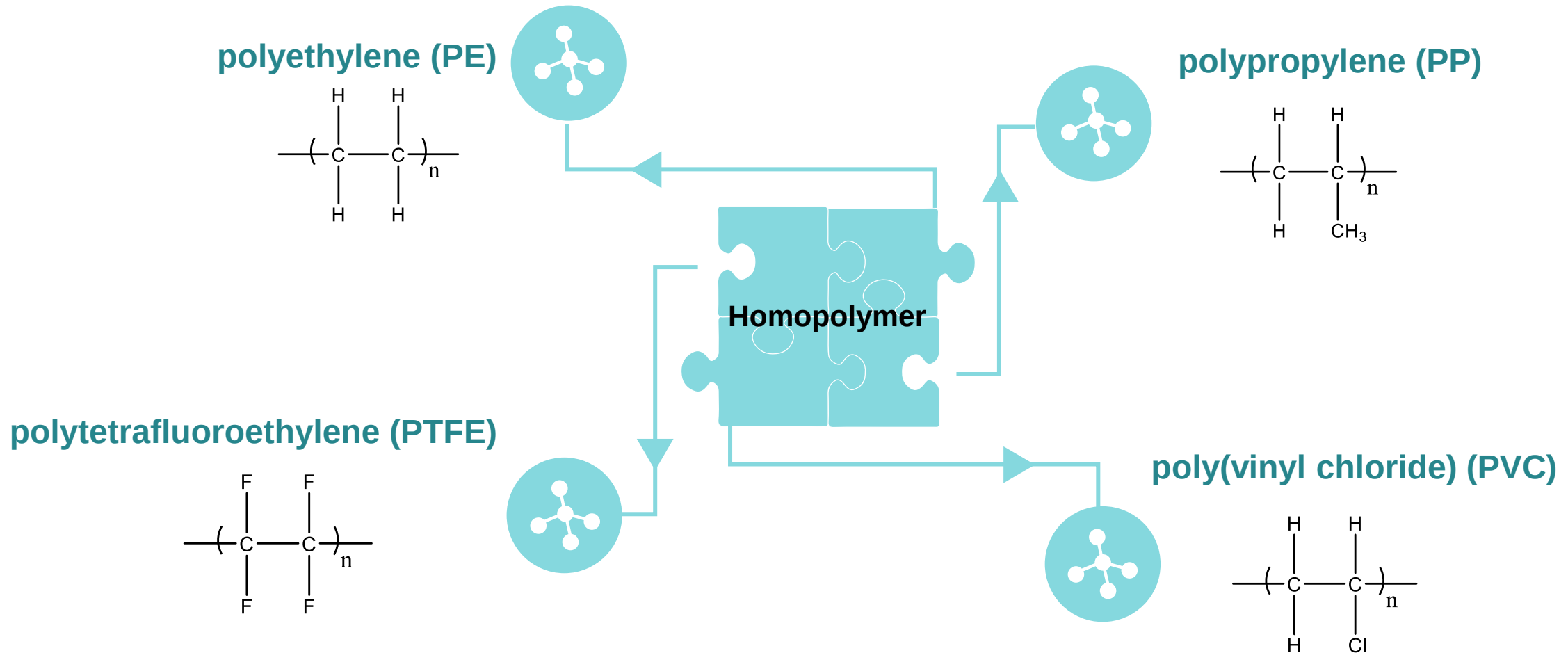


\*\* Crystalline polymers always contain some amorphous regions. \*\*

# Types of polymer



Only one species of monomer is used to build a macromolecule.



# Copolymer

At least two species of monomer are used to build a macromolecule.

## Block copolymer

compose of substantial sequences or blocks of each



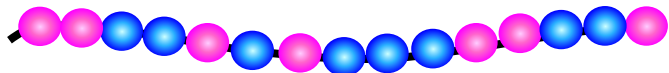
Block copolymer



Triblock copolymer

## Random copolymer

The distribution of the two monomers in the chain is essentially random.



01

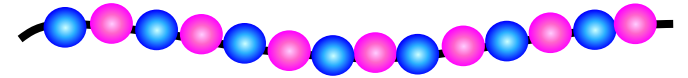
02

03

04

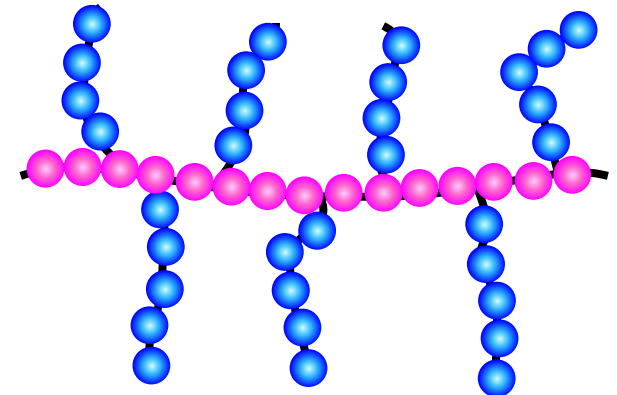
## Alternating copolymer

a regular placement along the chain

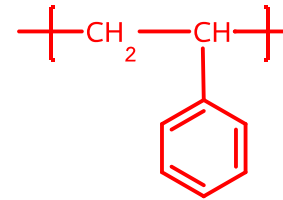
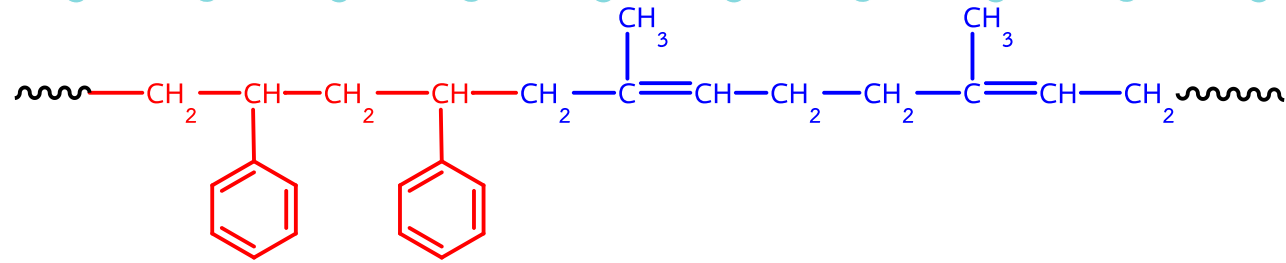


## Graft copolymer

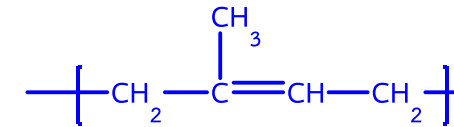
Blocks of one monomer are grafted on to a backbone of the other as branches.



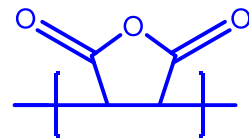
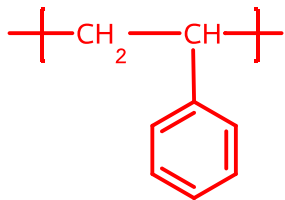
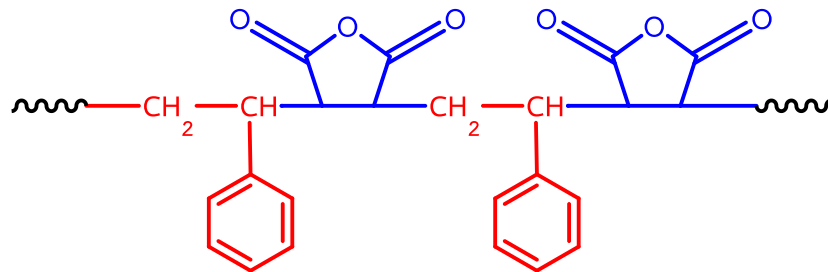
## Block copolymer



styrene repeating unit



isoprene repeating unit

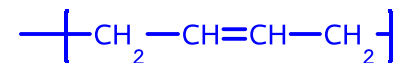
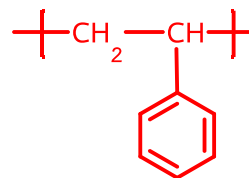
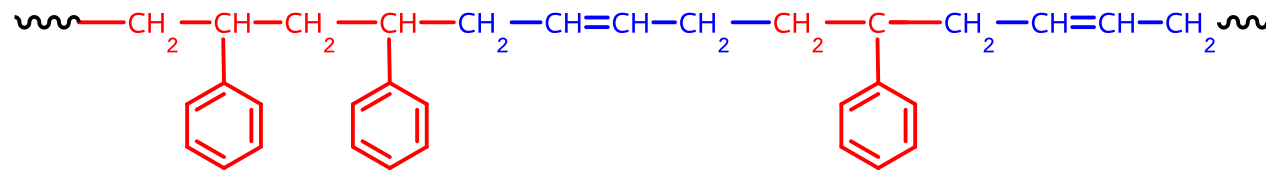


styrene repeating unit    maleic anhydride repeating unit

## Alternating copolymer

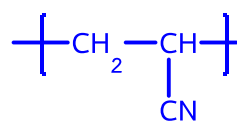
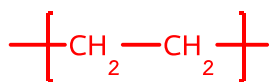
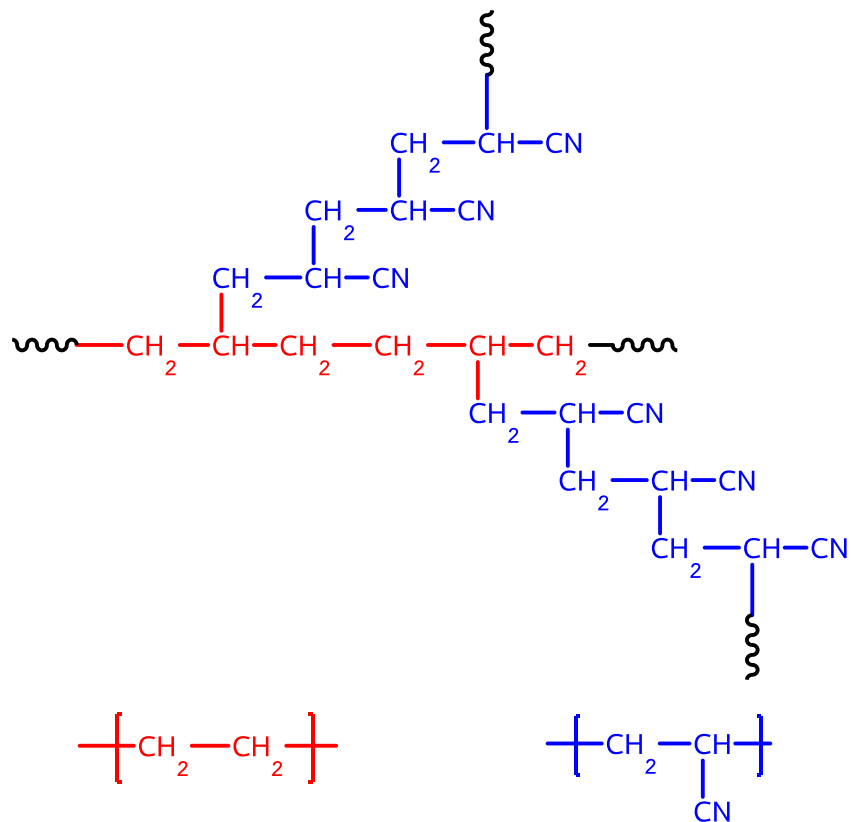


## Random copolymer



styrene repeating unit

butadiene repeating unit



ethylene repeating unit

acrylonitrile repeating unit

## Graft copolymer



# ***Nomenclature of copolymer***

***Unspecified***

poly(A-co-B)

***Block***

poly(A-b-B)

***Alternating***

poly(A-alt-B)

***Graft***

poly(A-g-B)

***Random***

poly(A-ran-B)

poly A-co-poly B

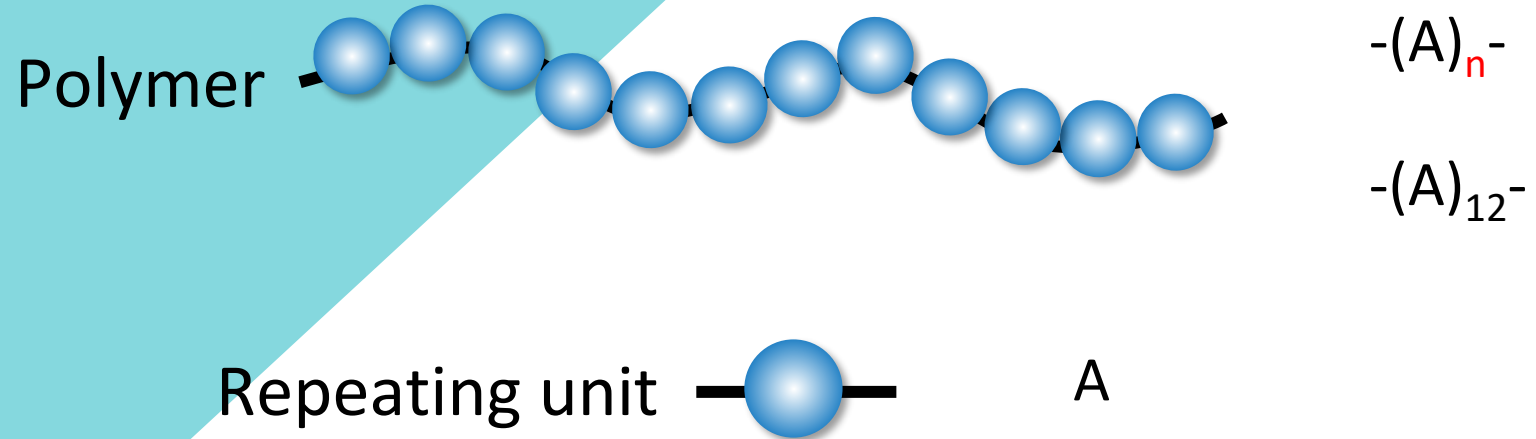
poly A-block-poly B

poly A-alt-poly B

poly A-graft-poly B

poly A-ran-poly B

# Degree of polymerization



Degree of polymerization or number of repeating unit (**n**)

$$DP_n = X_n = \frac{M_n}{M_o}$$

# Molecular weight determination

## Number average molecular weight

the statistical average molecular weight of all the polymer chains in the sample

$$\bar{M}_n = \frac{\sum N_i M_i}{\sum N_i} = \frac{\sum W_i}{\sum \frac{W_i}{M_i}} = \frac{1}{\sum \frac{w_i}{M_i}}$$

$$w_i = \frac{N_i M_i}{\sum N_i M_i}$$

measured by *colligative methods & end-group analysis*

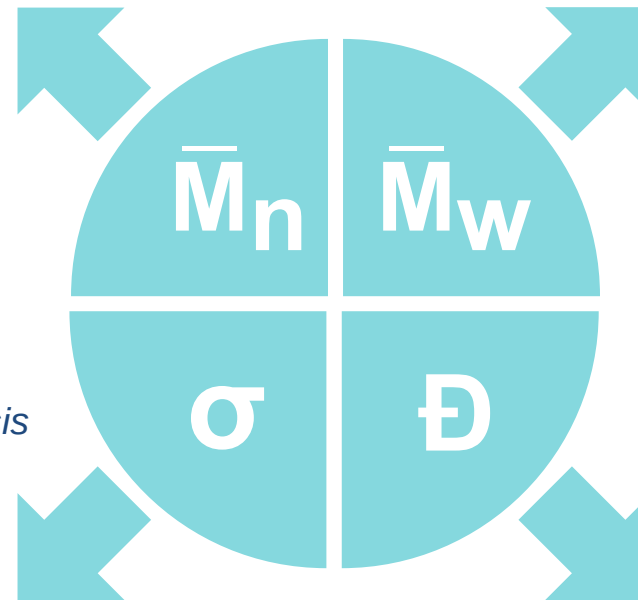
## Standard deviation of molecular weight

$$\sigma = \bar{M}_n \sqrt{\frac{\bar{M}_w}{\bar{M}_n} - 1}$$

## Weight average molecular weight

depends on the weight of each molecule

$$\bar{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{\sum W_i M_i}{\sum W_i} = \sum w_i M_i$$



$w_i$  is weight fraction of polymer with molecular weight  $M_i$

$W_i$  is weight of polymer with molecular weight  $M_i$

$N_i$  is number of mole of polymer with molecular weight  $M_i$

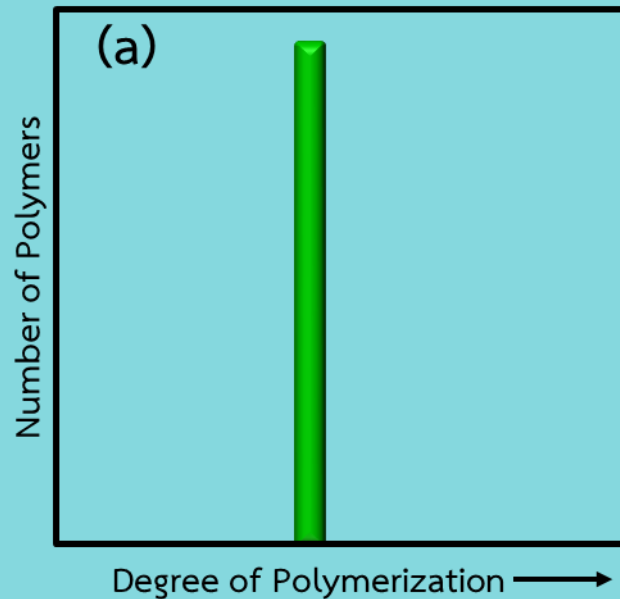
$M_i$  is the molecular weight of a chain with DP  $i$

## Polydispersity index

$$PDI = \frac{\bar{M}_w}{\bar{M}_n}$$

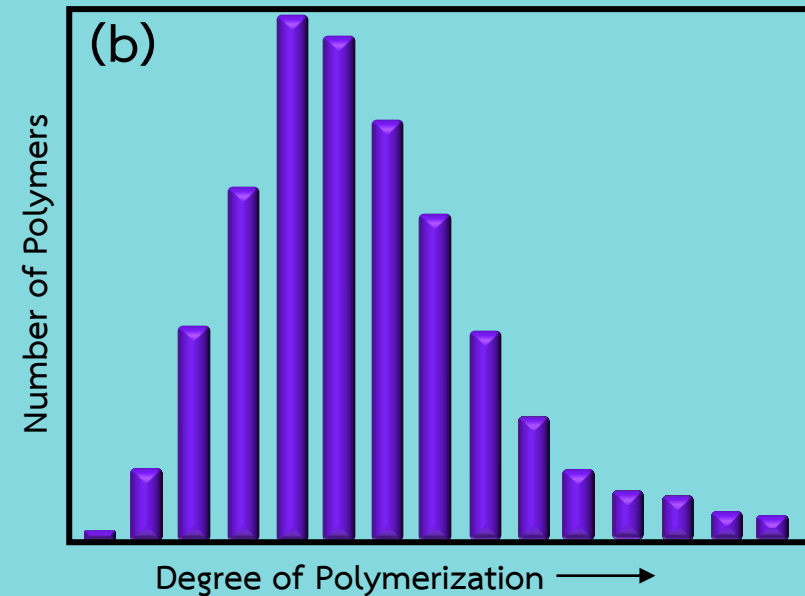
# Molecular weight distribution

**Polydispersity index (PDI,  $\bar{M}_w/\bar{M}_n$ ):** the distribution of molecular weight in a given polymer sample



**Monodisperse:  $PDI = 1$**

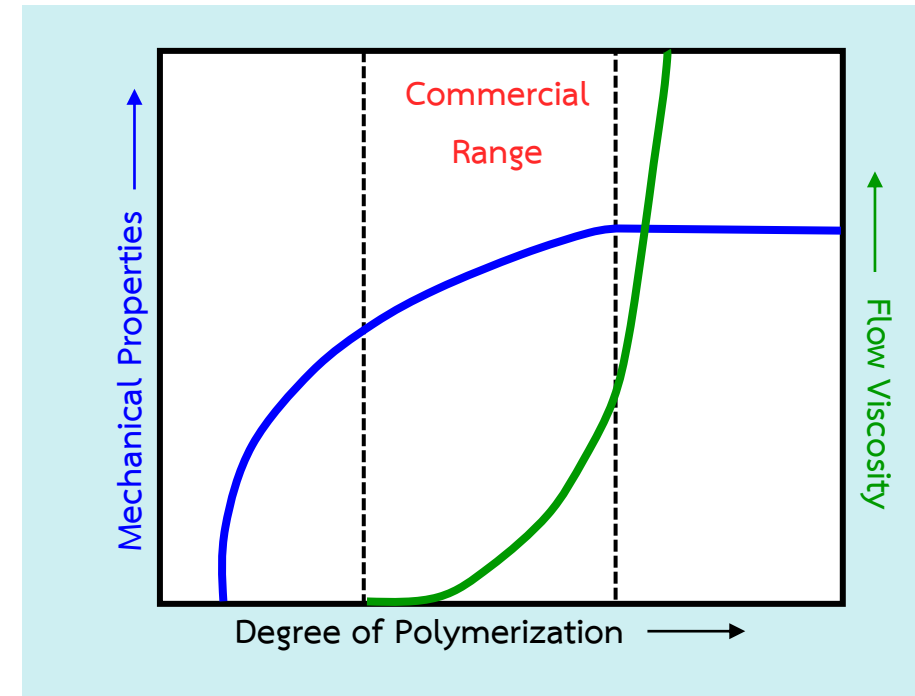
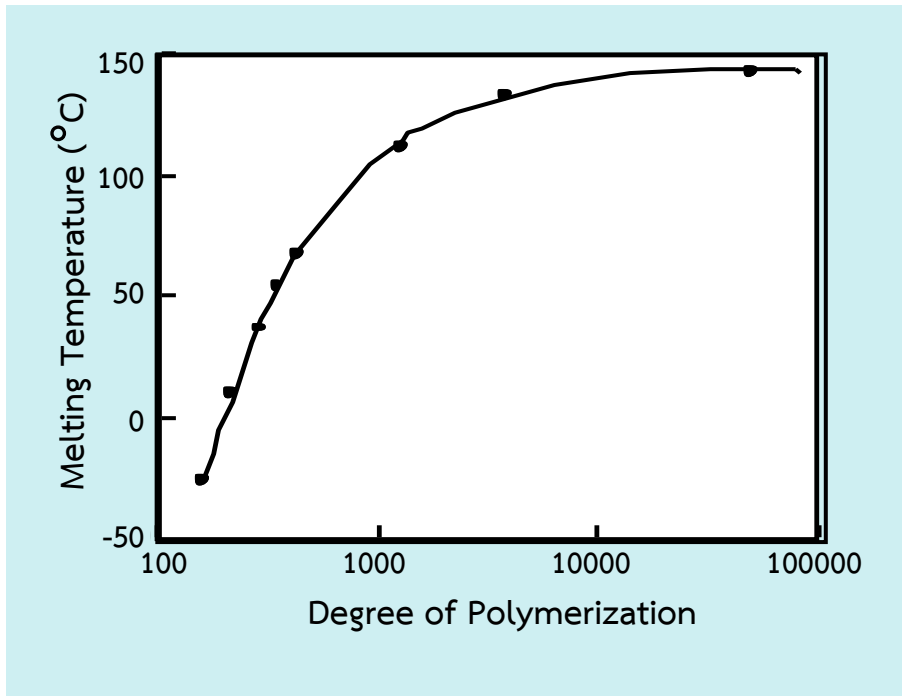
Polymer chains in the system  
have similar DP



**Polydisperse:  $PDI > 1$**

DP of polymer chains in the system  
is varied.

# Effect of molecular weight



## Calculation of average molecular weight

| Molecular weight<br>(g/mol) | Mole<br>(mol) |
|-----------------------------|---------------|
| 30,000                      | 9             |
| 50,000                      | 5             |

$$\bar{M}_n = \frac{\sum N_i M_i}{\sum N_i} = \frac{(9 \text{ mol} \times 30,000 \text{ g/mol}) + (5 \text{ mol} \times 50,000 \text{ g/mol})}{9 \text{ mol} + 5 \text{ mol}}$$

$$= 37,143 \text{ g/mol}$$

$$\bar{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i} = \frac{(9 \text{ mol} \times (30,000 \text{ g/mol})^2) + (5 \text{ mol} \times (50,000 \text{ g/mol})^2)}{(9 \text{ mol} \times 30,000 \text{ g/mol}) + (5 \text{ mol} \times 50,000 \text{ g/mol})}$$

$$= 39,615 \text{ g/mol}$$

$$\text{PDI} = \frac{\bar{M}_w}{\bar{M}_n} = \frac{39615 \text{ g/mol}}{37143 \text{ g/mol}} = 1.07$$

## Calculation of average molecular weight

|                                               | Mixture<br>A | Mixture<br>B |
|-----------------------------------------------|--------------|--------------|
| Molar mass of 1 <sup>st</sup> polymer (g/mol) | 1,000        | 1,000        |
| Weight fraction ( $w_i$ )                     | 0.5          | 0.4          |
| Molar mass of 2 <sup>nd</sup> polymer (g/mol) | 25,000       | 6,000        |
| Weight fraction ( $w_i$ )                     | 0.5          | 0.6          |
| $M_n$ (g/mol)                                 |              |              |
| $M_w$ (g/mol)                                 |              |              |
| PDI                                           |              |              |



# 342-201

## Chapter 2: Polymerization

Assoc. Prof. Dr. Sirinya Chantarak



# TOPICS

**01**

**Step-growth Polymerization**

**02**

**Free-radical Polymerization**

**03**

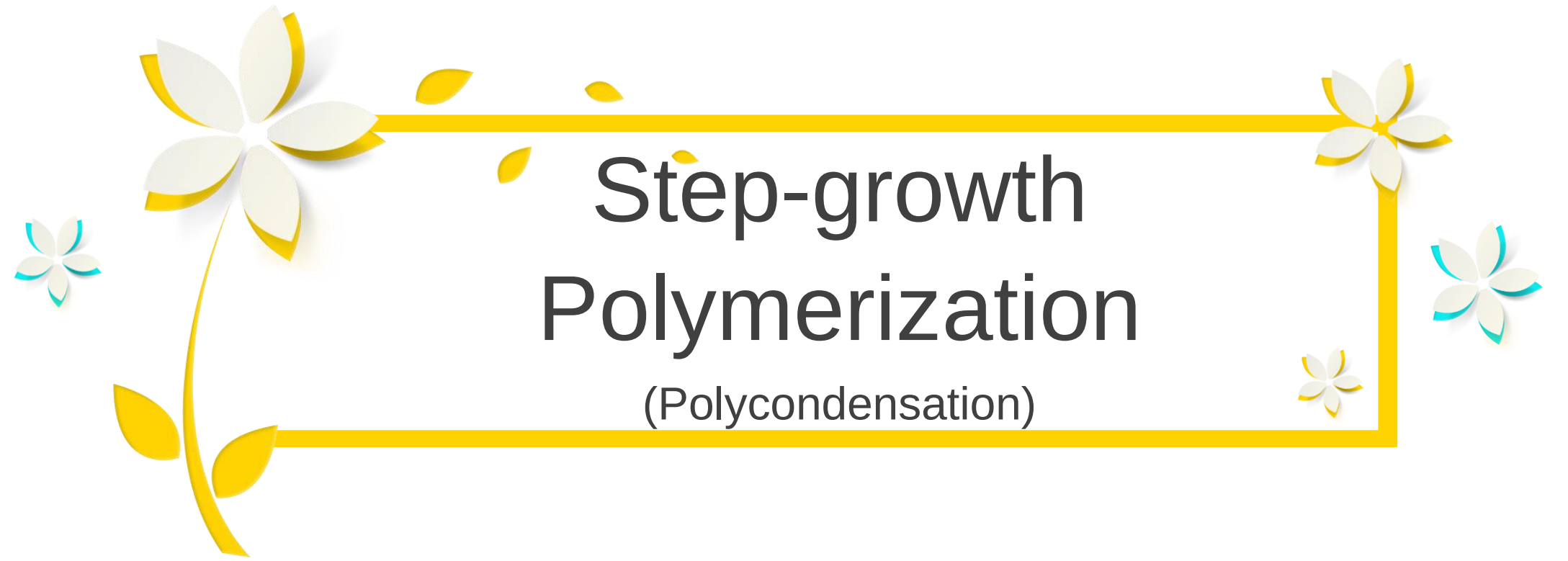
**Cationic Polymerization**

**04**

**Anionic Polymerization**

**05**

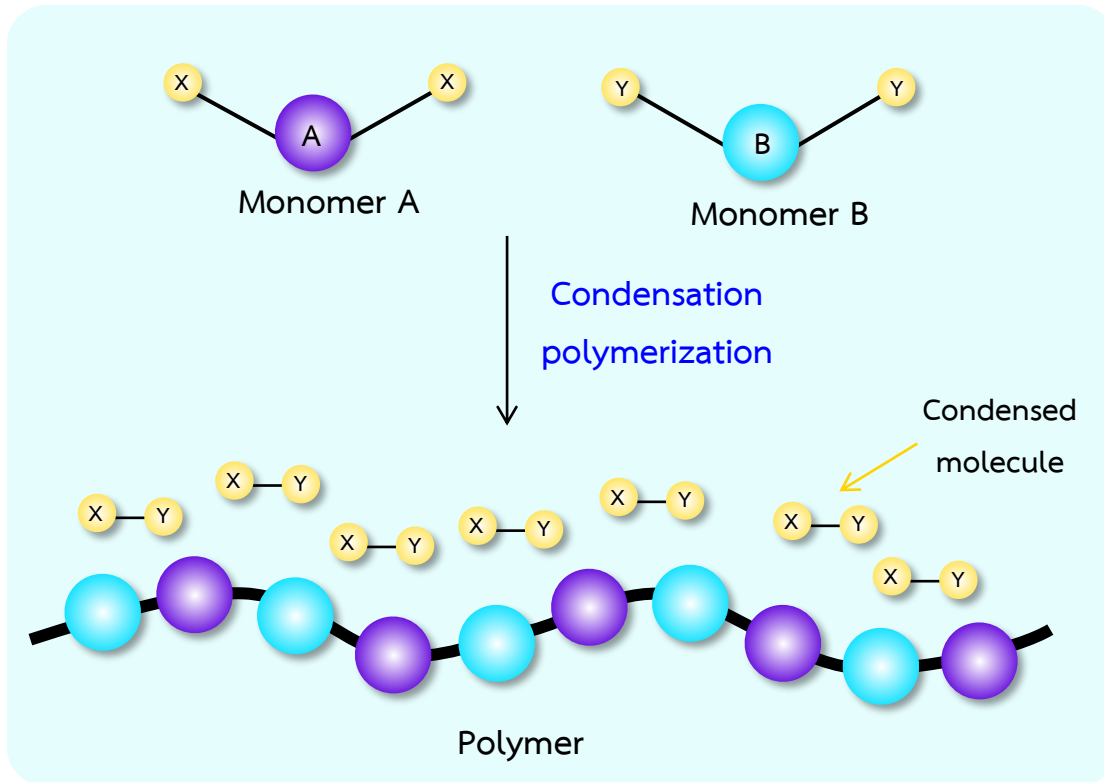
**Coordination Polymerization**



# Step-growth Polymerization

(Polycondensation)

# Step-growth polymerization



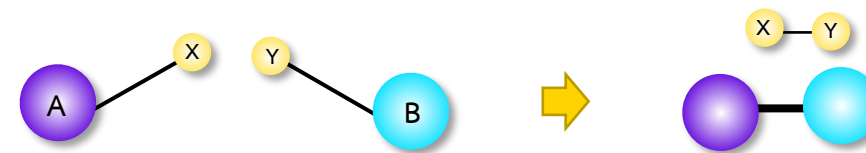
|         |   |         |   |          |
|---------|---|---------|---|----------|
| monomer | + | monomer | = | dimer    |
| dimer   | + | monomer | = | trimer   |
| dimer   | + | dimer   | = | tetramer |
| trimer  | + | monomer | = | tetramer |
| trimer  | + | dimer   | = | pentamer |
| trimer  | + | trimer  | = | hexamer  |



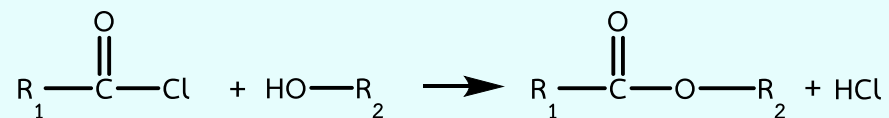
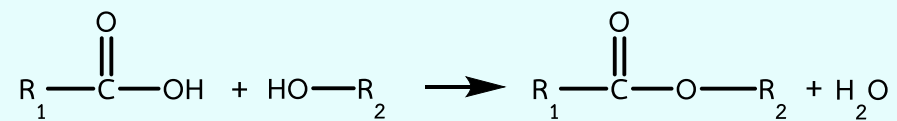
Random reaction of two **difunctional monomers**, such as **carboxylic acid** or **hydroxyl groups**.

Step-Growth polymerization occurs with the elimination of a **small molecule**, e.g., water.

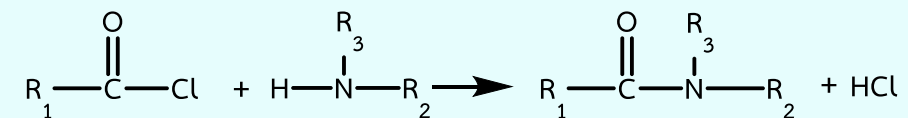
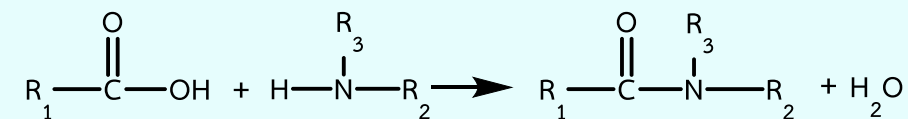
# Basic condensation reaction



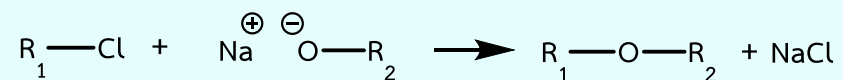
## Esterification



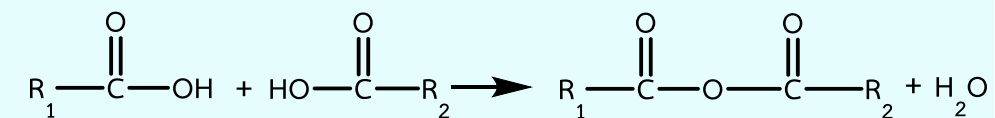
## Amide formation



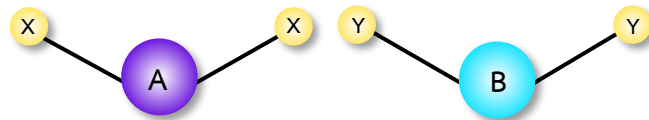
## Ether formation



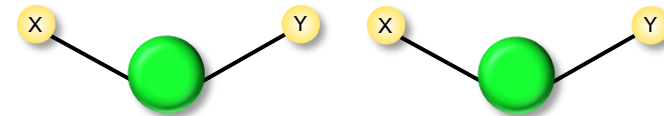
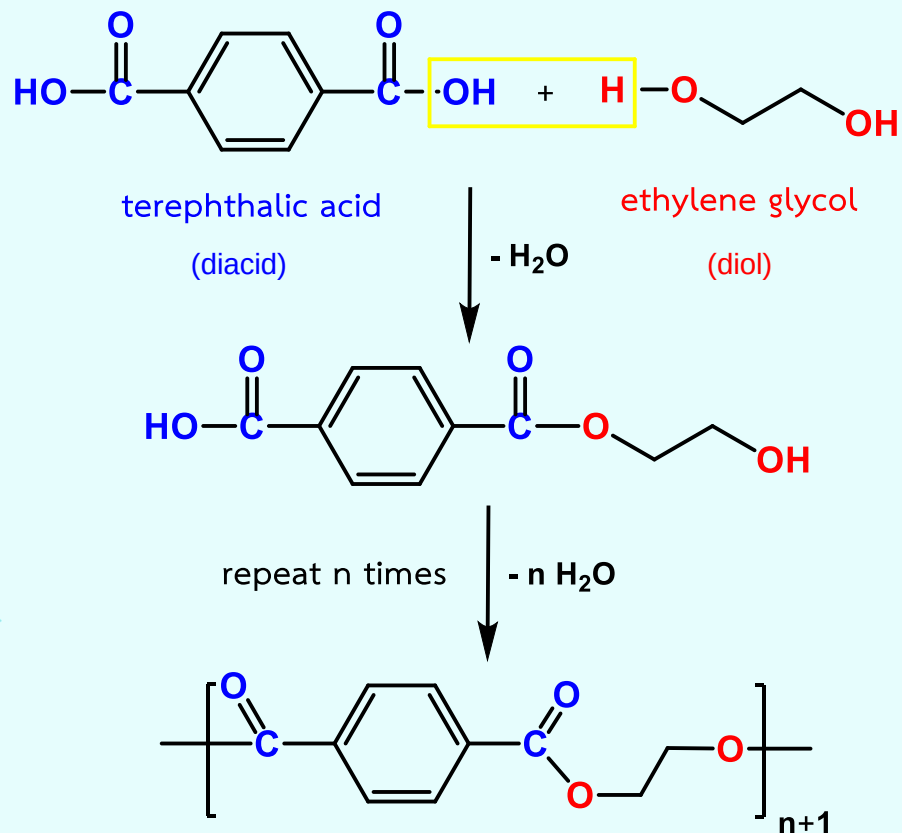
## Anhydride formation



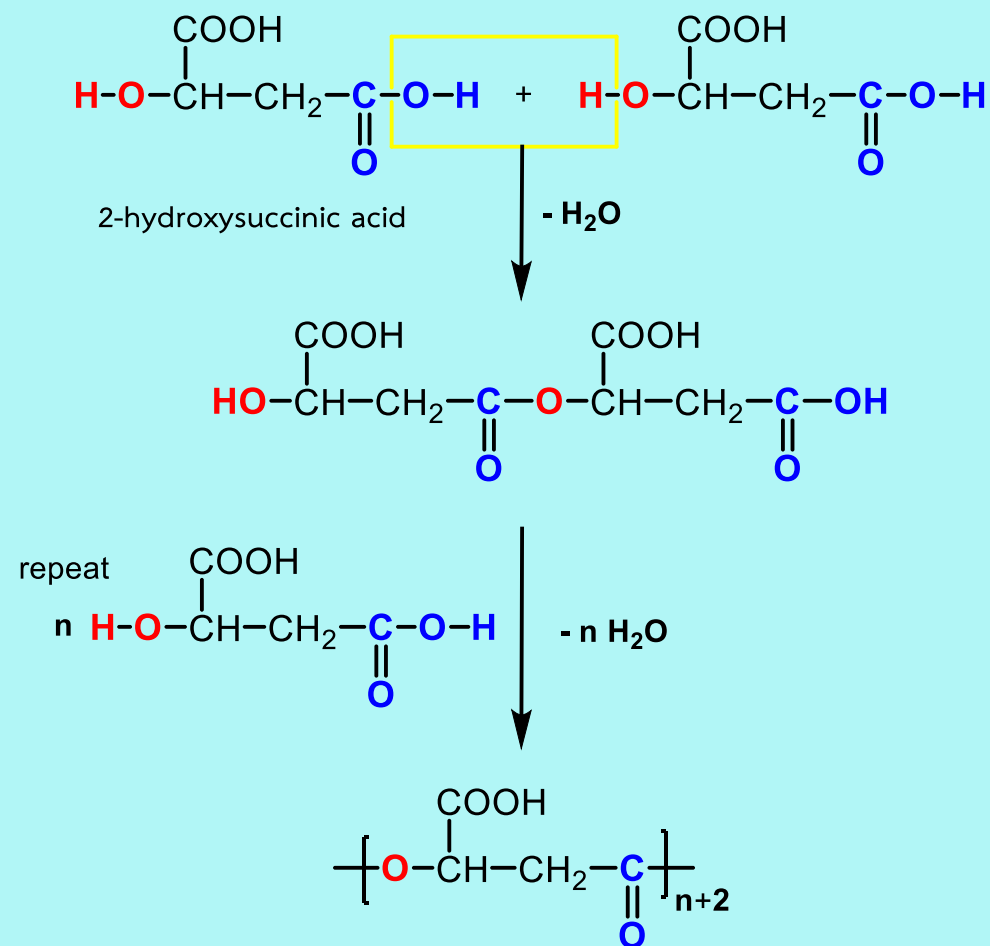
# Polyester



difunctional monomer / homofunctional monomer

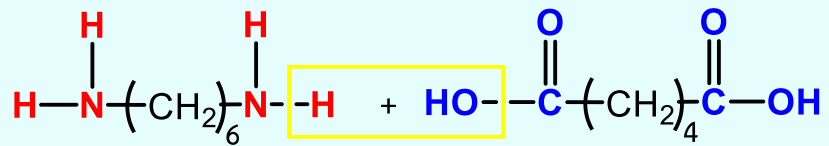
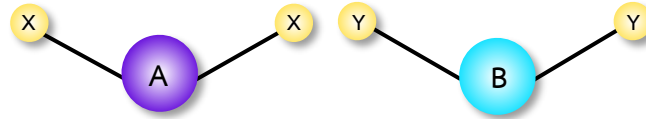


bifunctional monomer / heterofunctional monomer



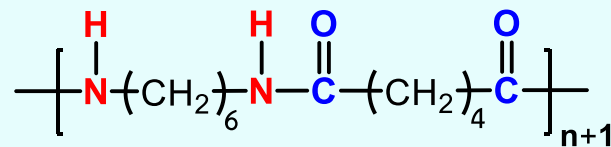
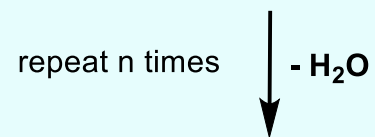
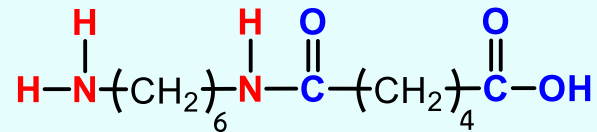
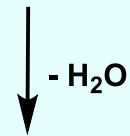
# Polyamide

difunctional monomer / homofunctional monomer



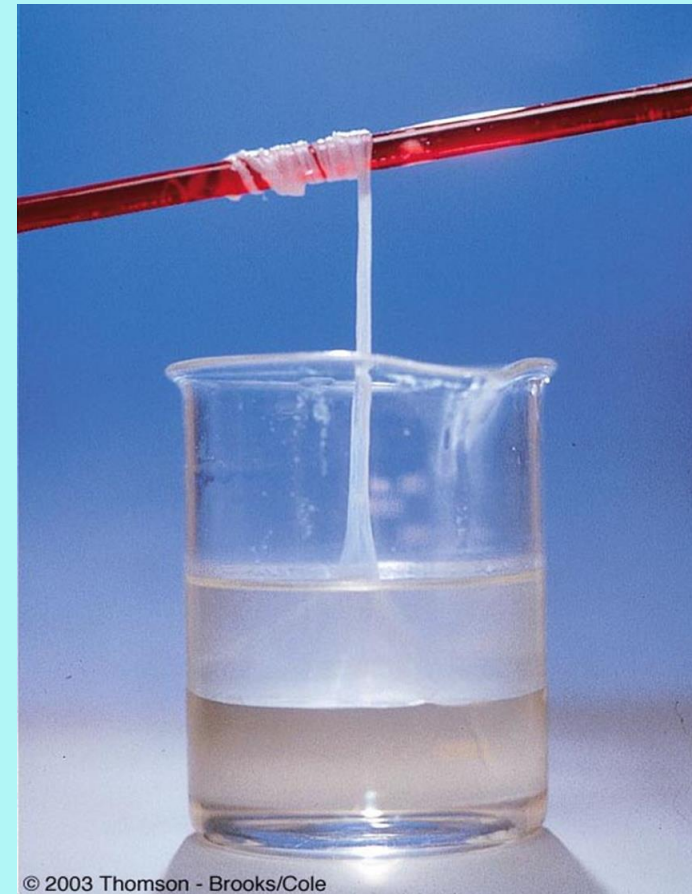
hexamethylene diamine  
(diamine)

adipic acid  
(dicarboxylic acid)



Nylon 6,6

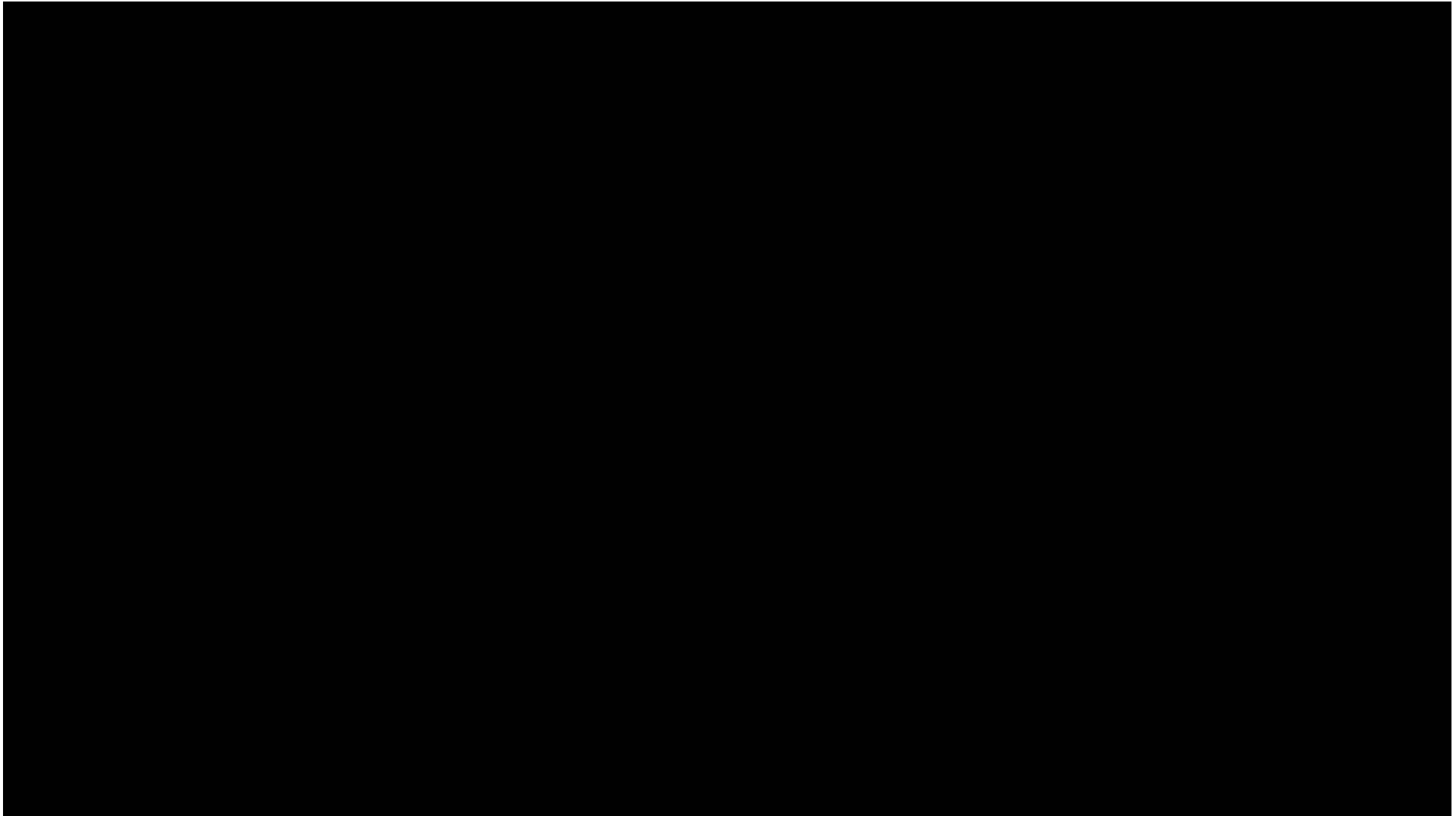
## Interfacial polymerization



© 2003 Thomson - Brooks/Cole

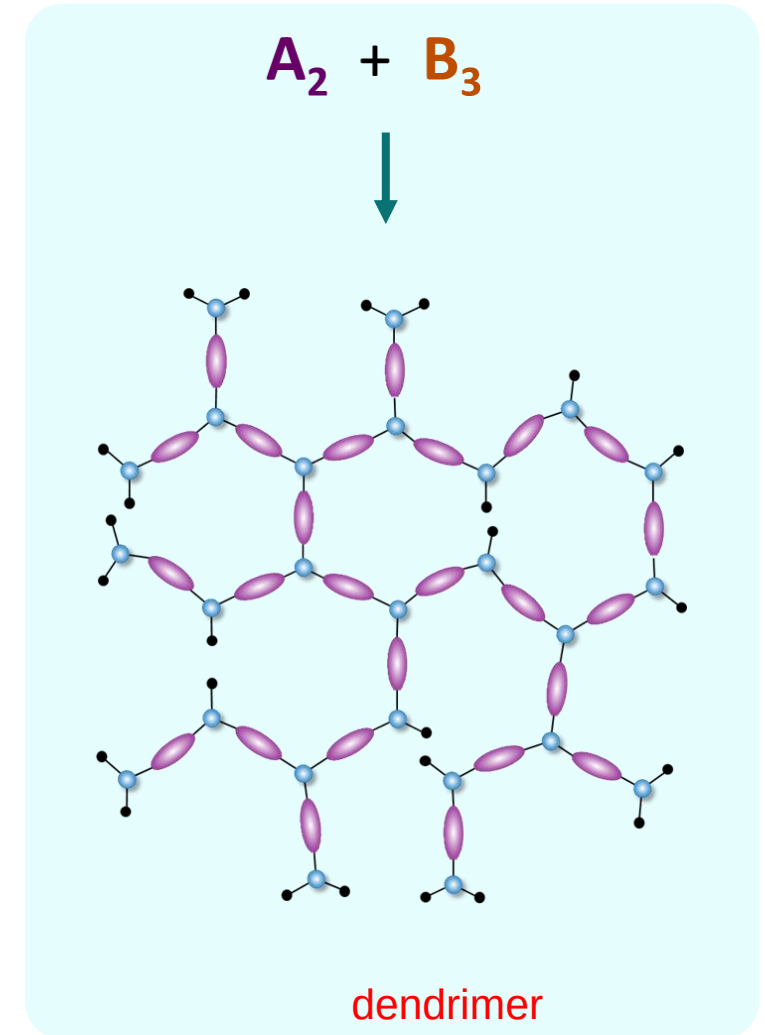
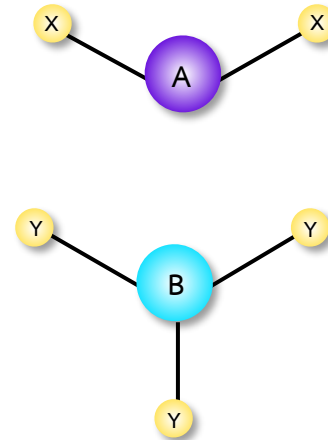
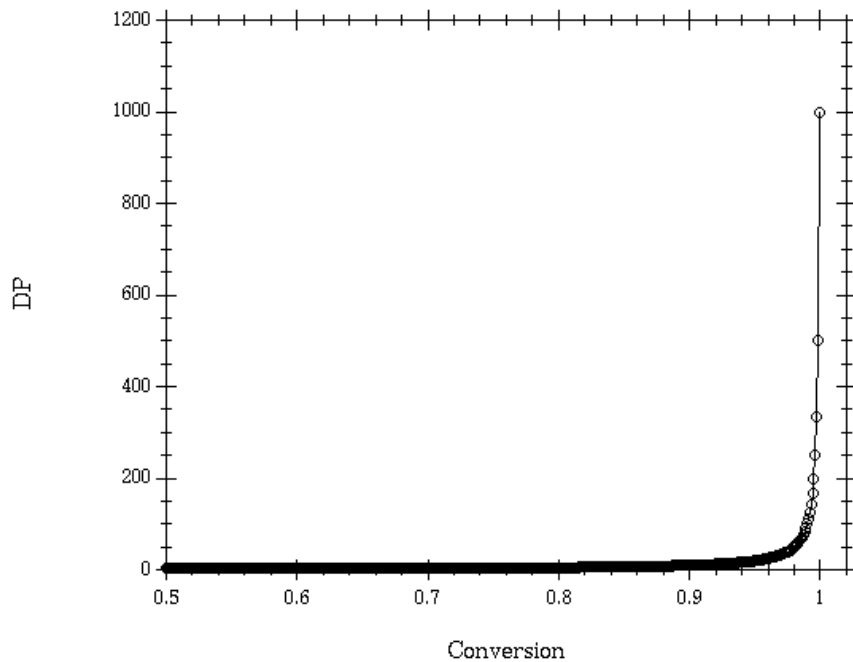


# Interfacial polymerization



# Characteristics of step-growth polymerization

- ❁ Reaction is slow and often requires heat.
- ❁ They are generally lower in molecular weight.
- ❁ The terminal functional groups on a chain remain active.
- ❁ High molecular weight are obtained at the end of the reaction.



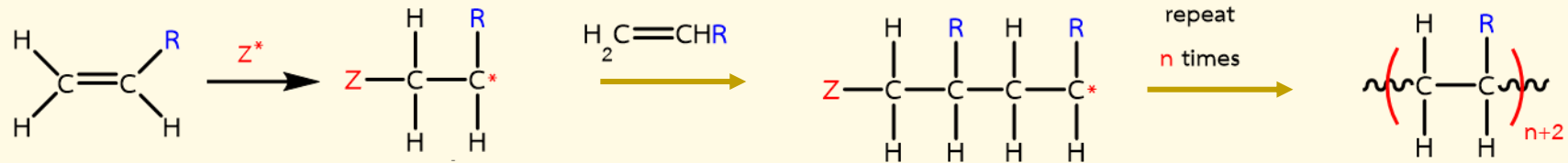
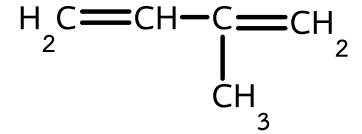
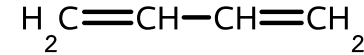
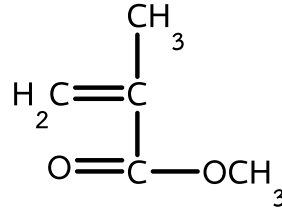
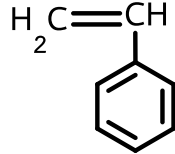
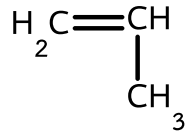
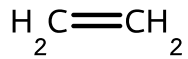


# Chain-growth Polymerization

(Addition polymerization)

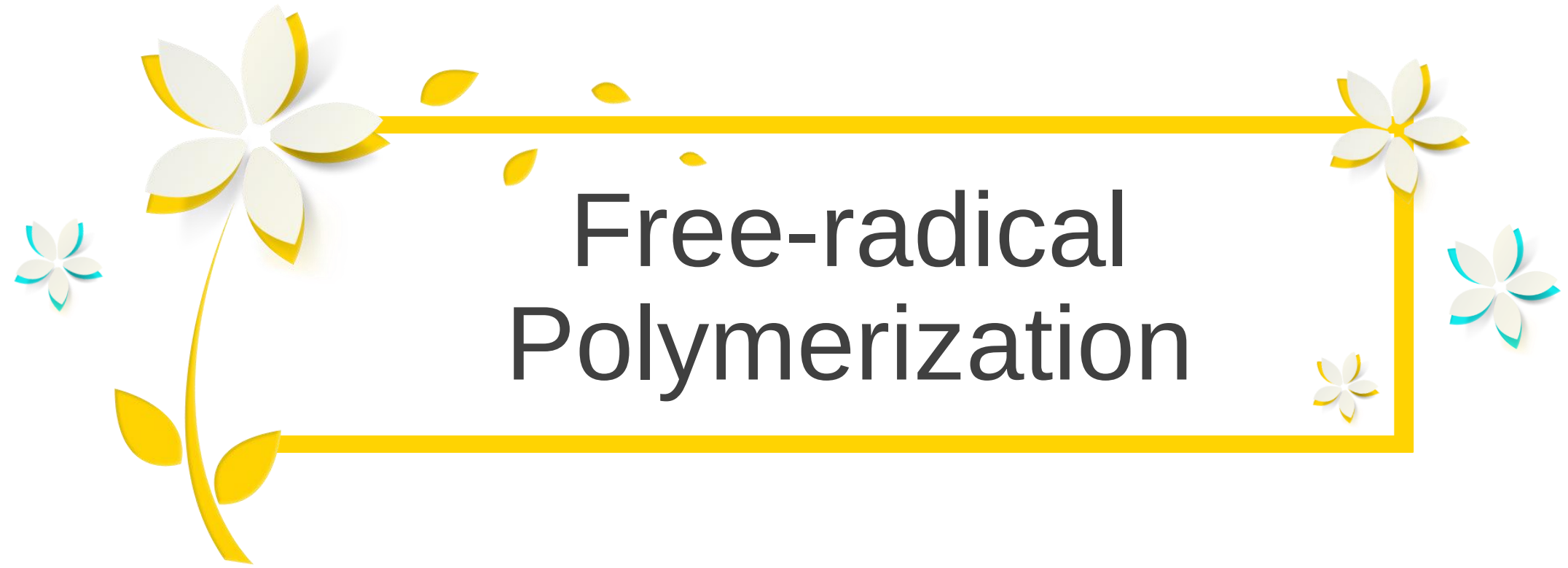
# Chain-growth polymerization

## Vinyl monomer / diene monomer



$\text{Z}^*$  is an initiating species (free radical, cation, or anion).

The polymerization may occur via any of a **free radical mechanism**, a **cationic**, or **anionic** mechanism, or a **co-ordination** catalyst mechanism.

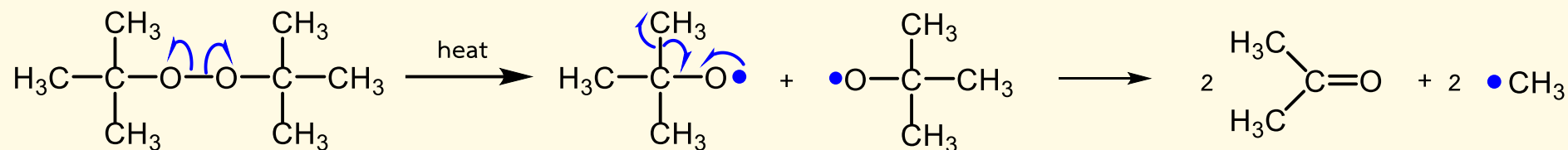


# Free-radical Polymerization

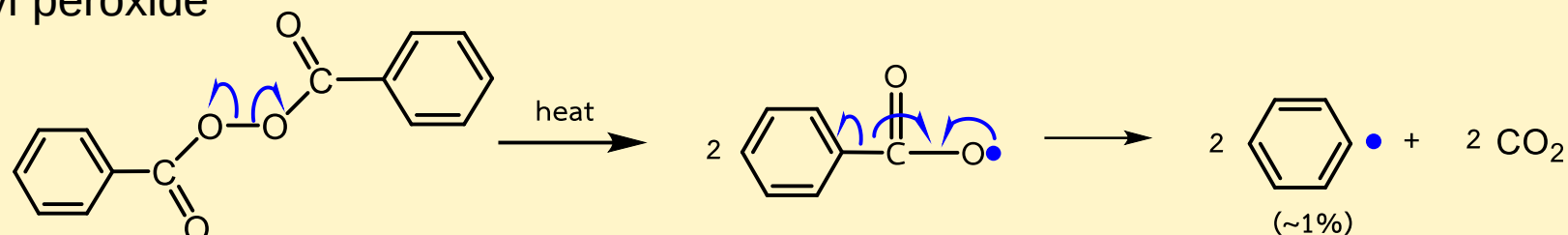
# Free-radical polymerization

**Radical initiator:** peroxide or certain azo compounds

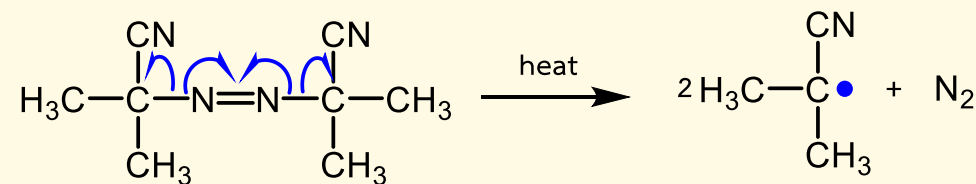
🌸 *tert*-butyl peroxide



🌸 benzoyl peroxide



🌸 azobisisobutyronitrile

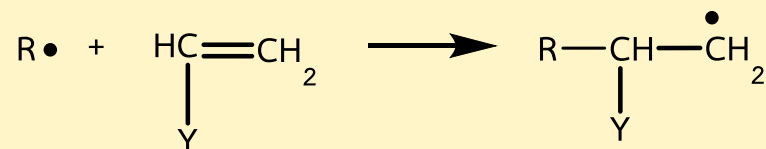
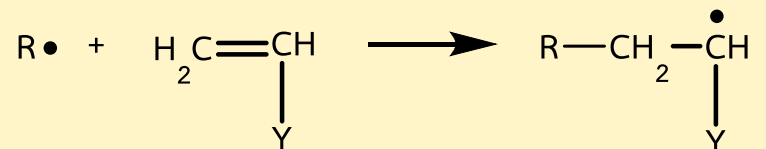


# Steps of reaction

## 1. Initiator-fragmentation step

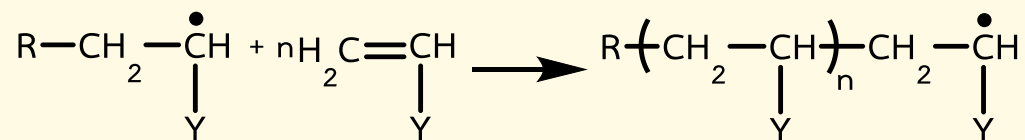


## 2. Initiation step



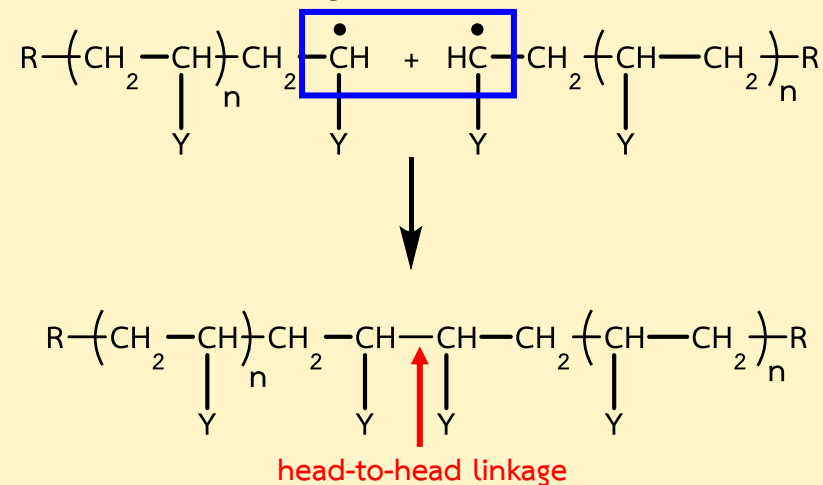
carbon radical

## 3. Propagation step

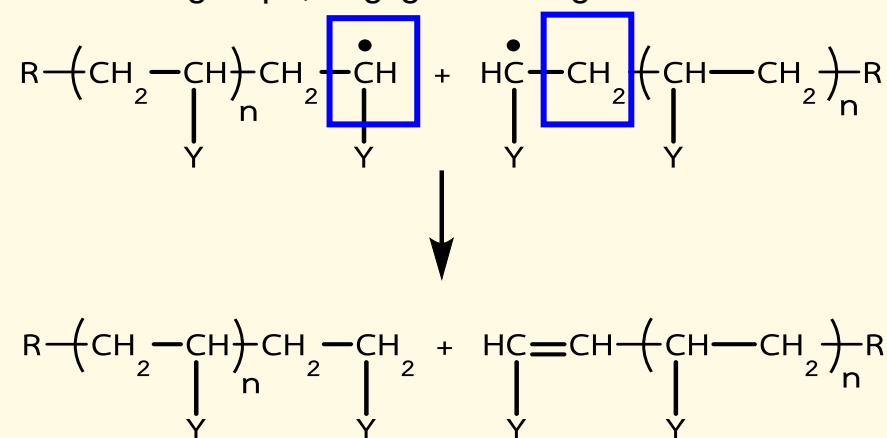


## 4. Termination step

 **combination:** creates one head-to-head link, and double molecular weight.

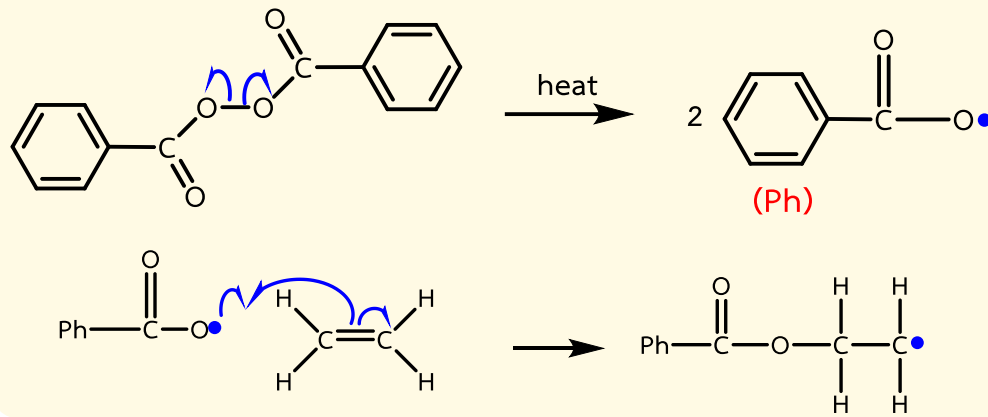


 **disproportionation:** two chains are terminated but with different end groups, negligible change to MW.

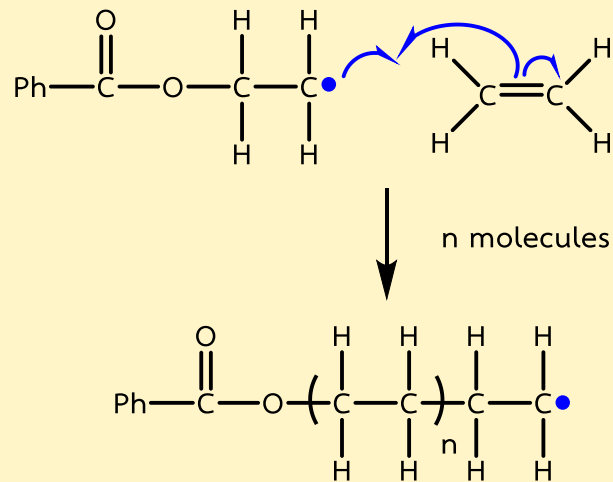


# Polymerization of polyethylene

## Initiation step

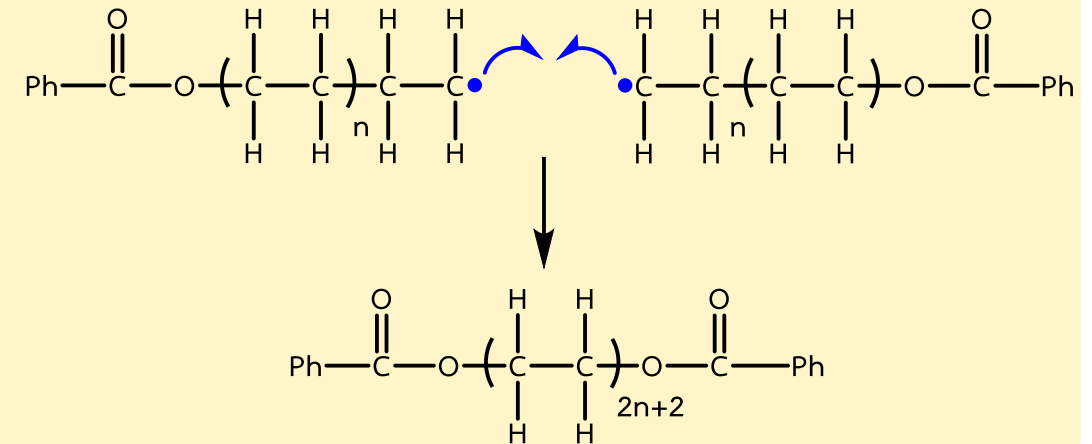


## Propagation step

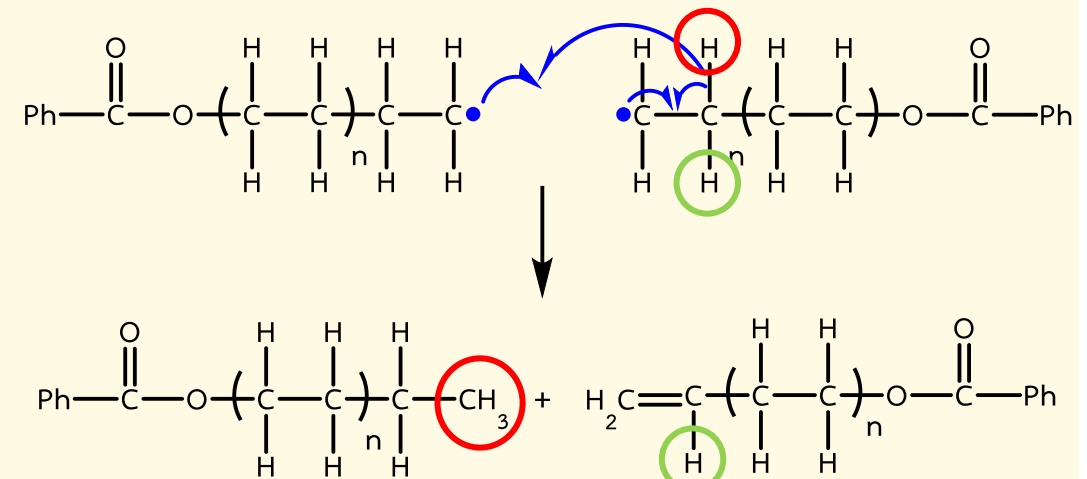


## Termination step

### (1) Combination

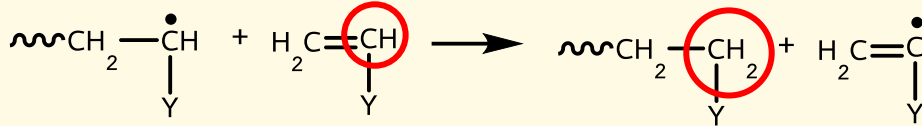


### (2) Disproportionation

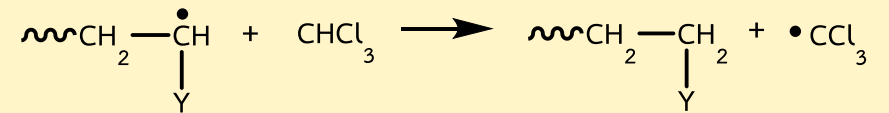


# Chain transfer

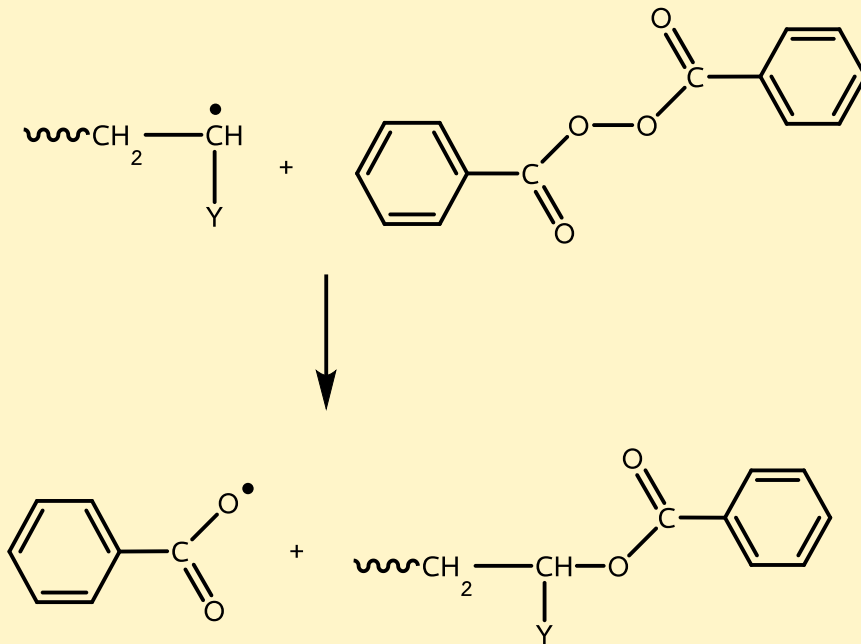
## Chain-transfer to monomer



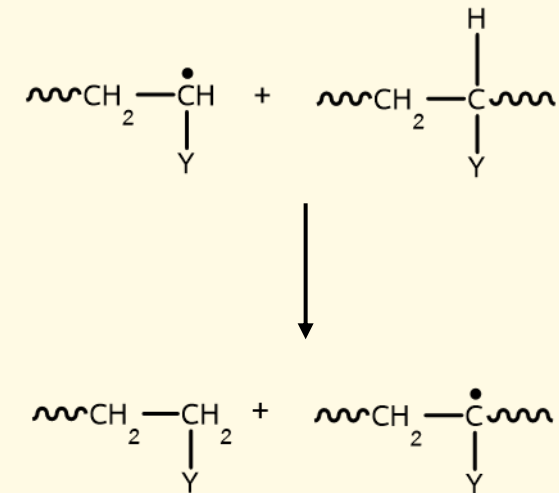
## Chain-transfer to solvent



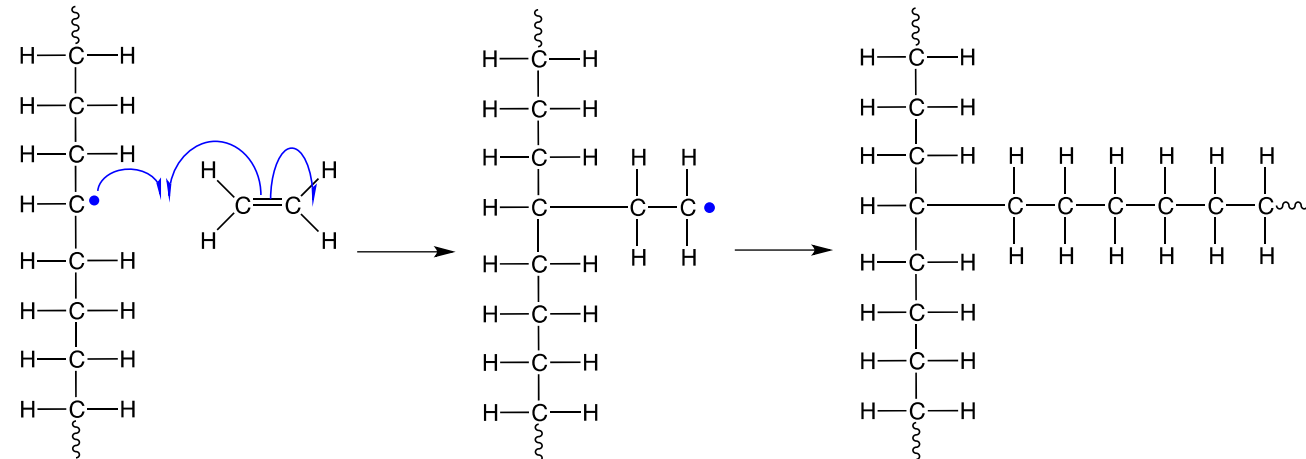
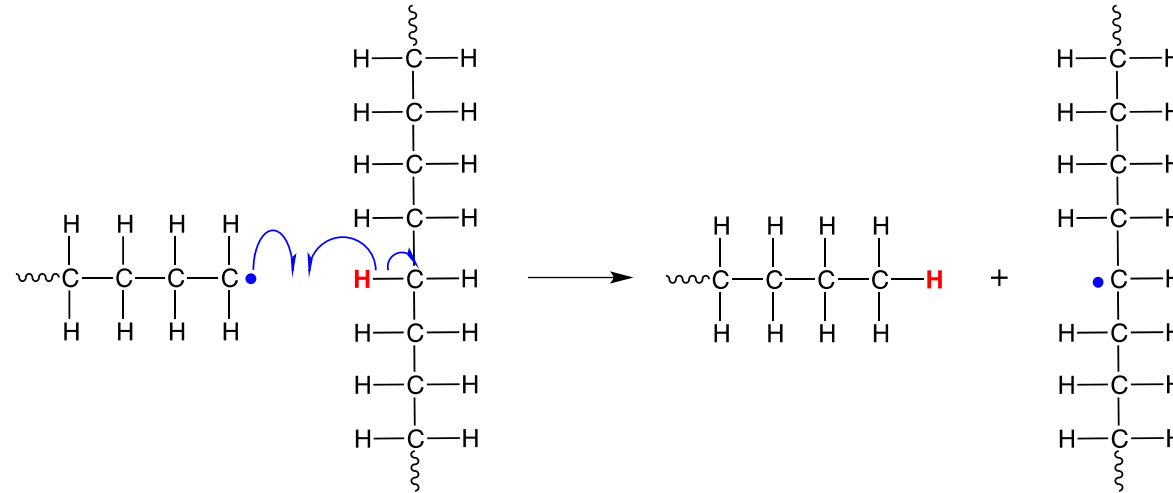
## Chain-transfer to initiator



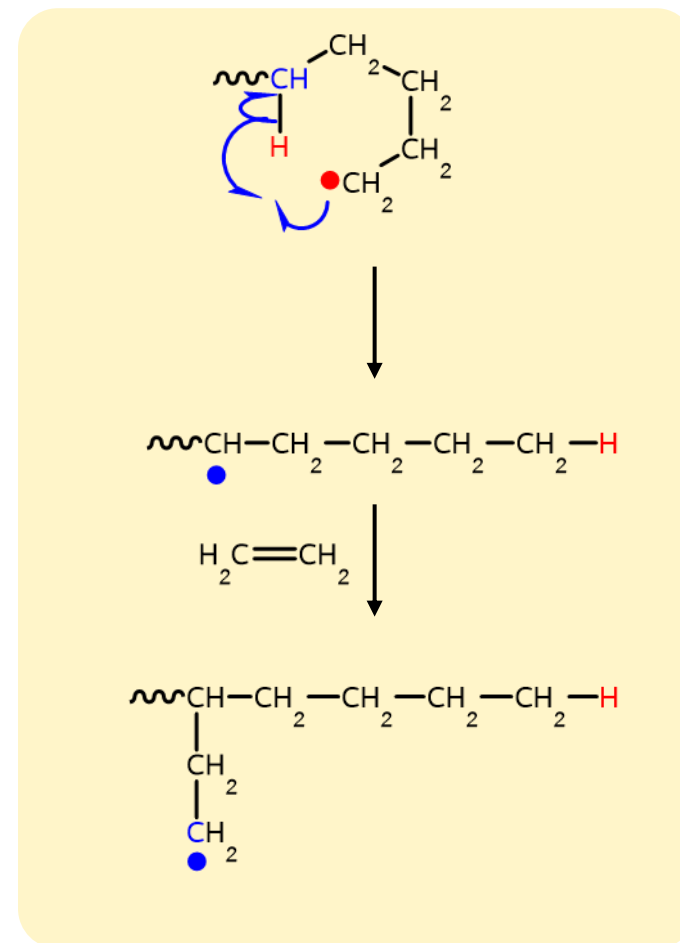
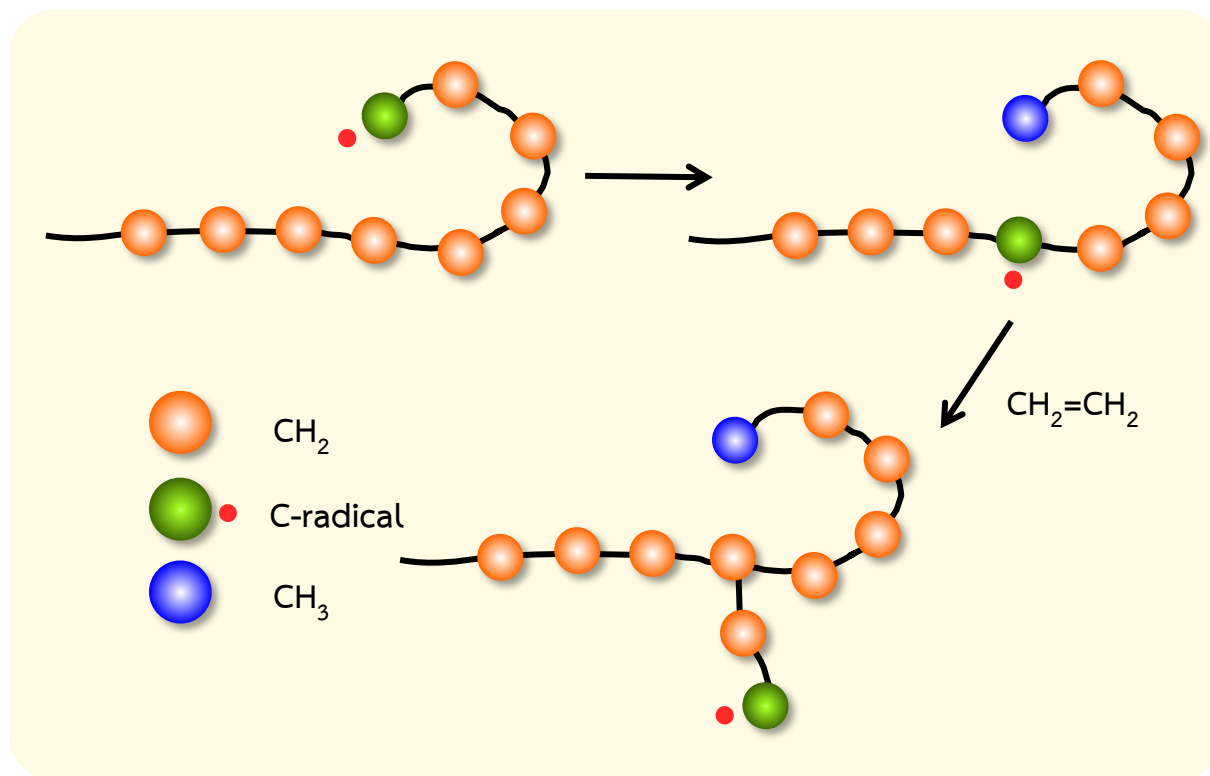
## Chain-transfer to polymer



❁ Chain transfer to polymer followed by chain propagation results in branched polymer.

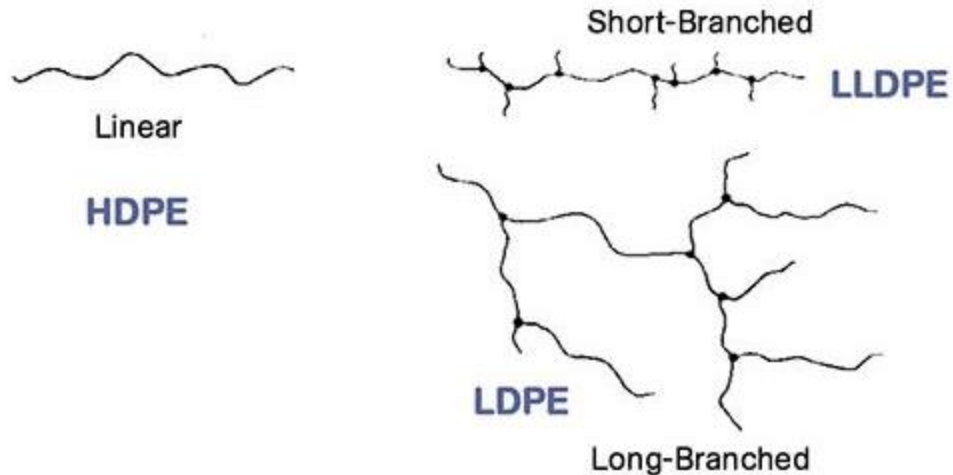


# Back-biting: intramolecular hydrogen subtraction



# Effect of branching on density

❁ Long branches or side chains keep the molecules apart and creates the low density.



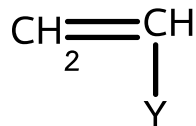
|                              | LDPE                                                                           | HDPE                                                                          |
|------------------------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Density (g/cm <sup>3</sup> ) | 0.91-0.93                                                                      | 0.94-0.96                                                                     |
| Crystallinity (%)            | 40-60                                                                          | 70-90                                                                         |
| Melting Temp. (°C)           | 105-115                                                                        | 113-138                                                                       |
| Properties                   | Soft and flexible<br>Transparent<br>Chemically stable<br>High tensile strength | Stiff and hard<br>Opaque<br>More chemically stable<br>Higher tensile strength |
| Uses                         | plastic bags<br>plastic wrap<br>squeeze bottles                                | bottles<br>buckets, containers<br>toys                                        |





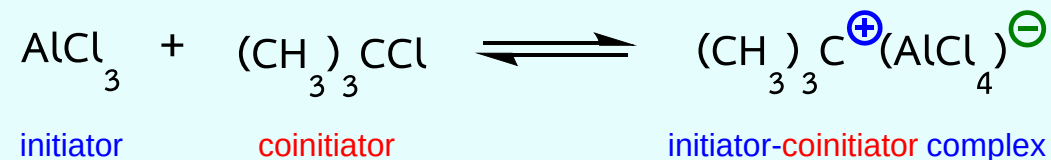
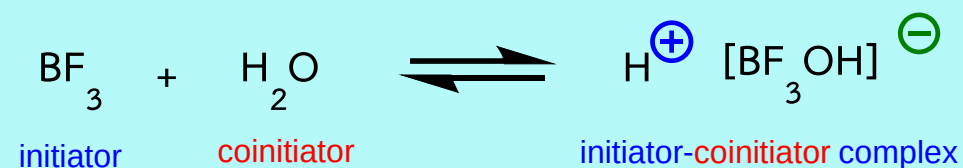
# Cationic Polymerization

# Cationic polymerization



Y = e<sup>-</sup> donating group (-CH<sub>3</sub>, OCH<sub>3</sub>, -Ph)

- ❁ The initiator is an acid.
- ❁ Protonic acids: strong acids H<sub>2</sub>SO<sub>4</sub>, HClO<sub>4</sub>
- ❁ Lewis acids: metal halides (AlCl<sub>3</sub>, BF<sub>3</sub>, SnCl<sub>4</sub>, SbCl<sub>5</sub>, ZnCl<sub>2</sub>, TiCl<sub>4</sub>) and organometallic derivatives (RAlCl<sub>2</sub>, R<sub>2</sub>AlCl, R<sub>3</sub>Cl)

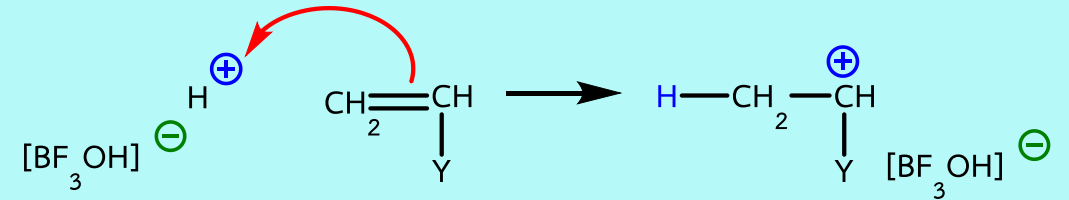
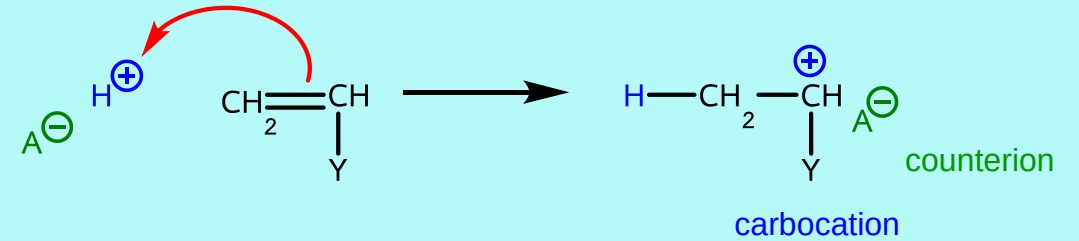


# Steps of reaction

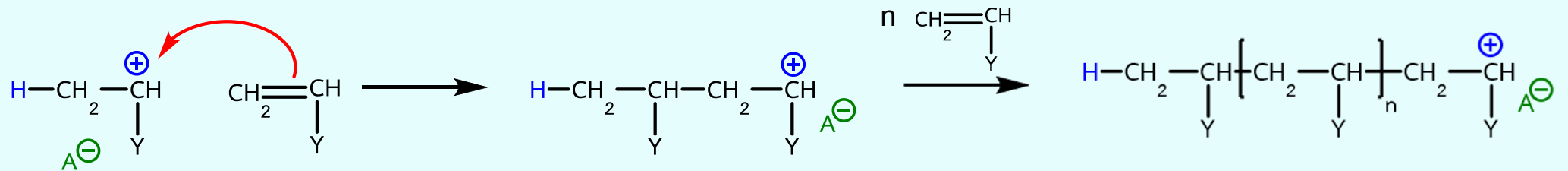
## 1. Initiator-fragmentation step



## 2. Initiation step

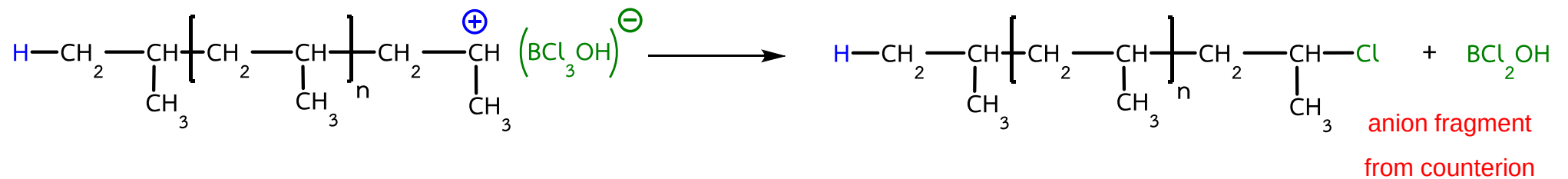
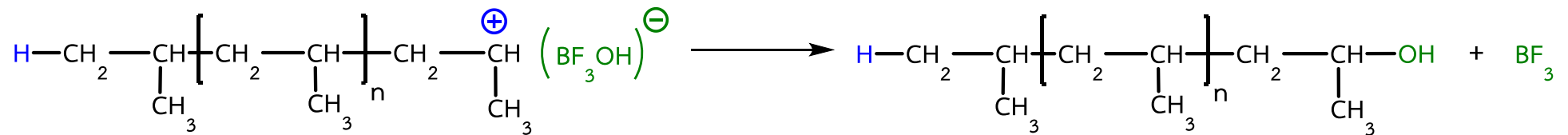
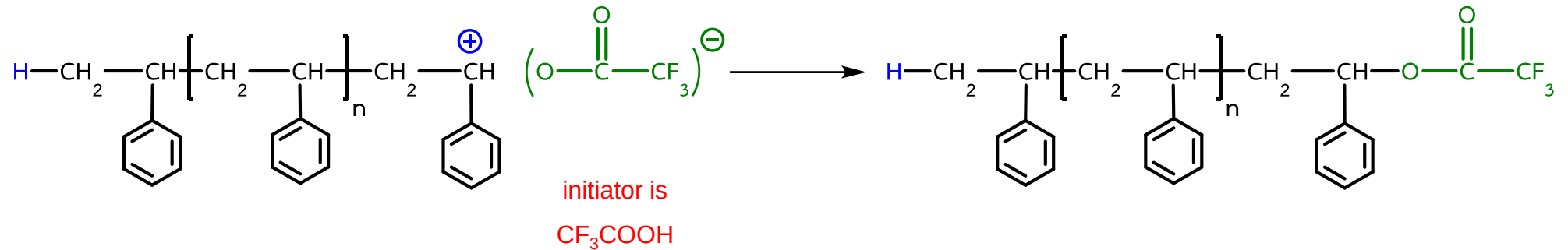


## 3. Propagation step

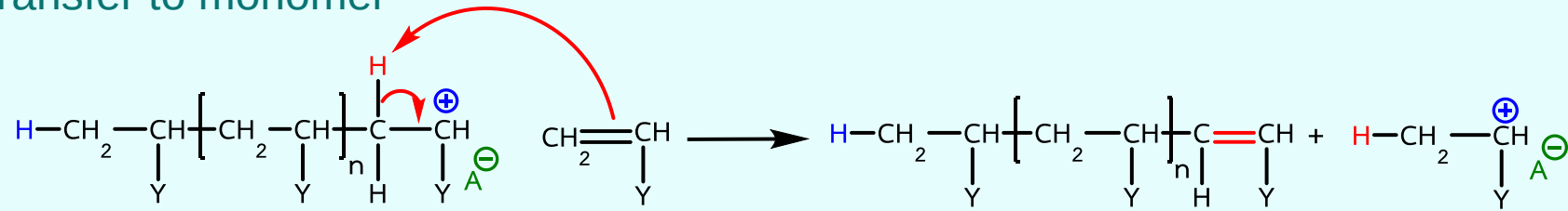


## 4. Termination step

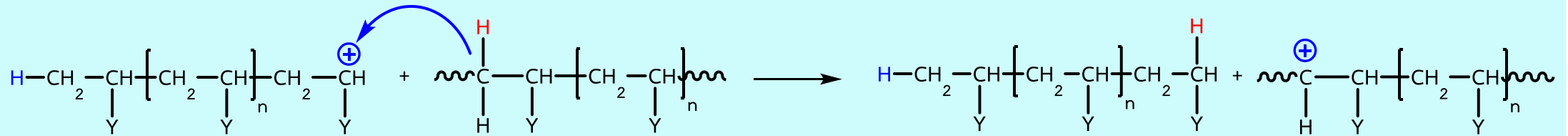
### (a) Combination with counterion



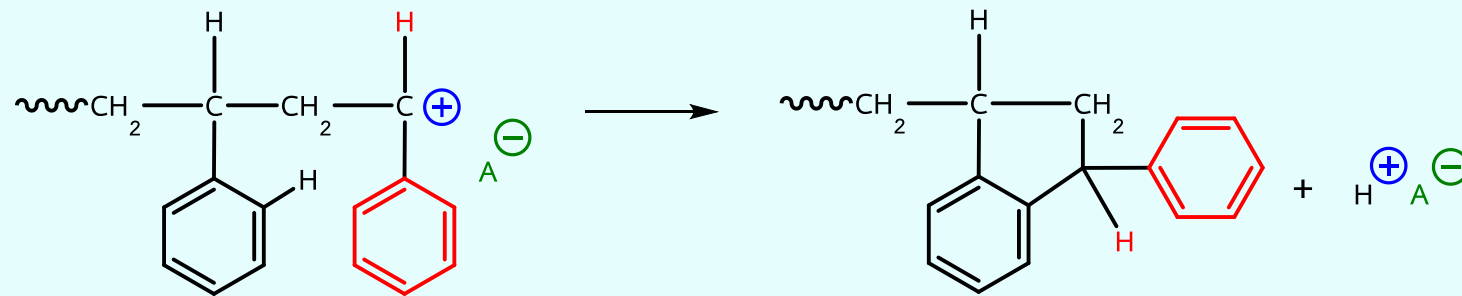
(b) chain transfer to monomer



(c) chain transfer to polymer



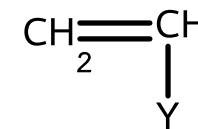
(d) back biting



# Electron donating groups

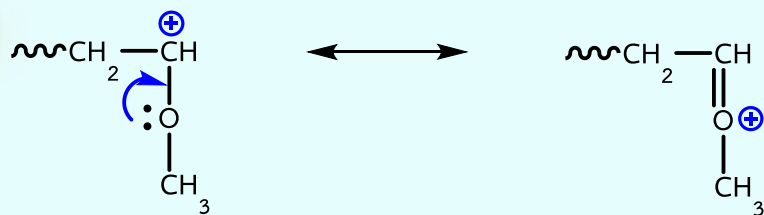
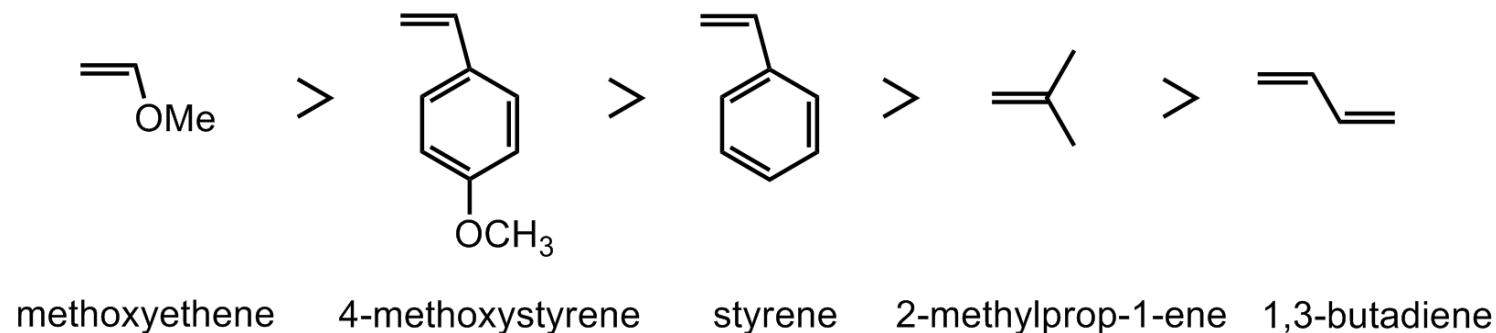
✿ The R groups can stabilize the propagating species by resonance.

✿ The propagating site of reactivity is a **carbocation**.

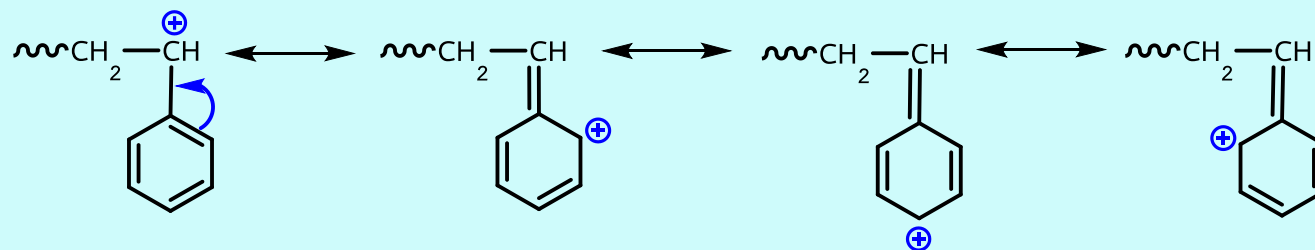


Y = e<sup>-</sup> donating group

Decreasing reactivity of  
olefin monomers



resonance isomers of methoxyethene

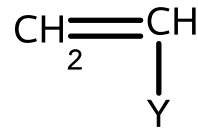


resonance isomers of styrene



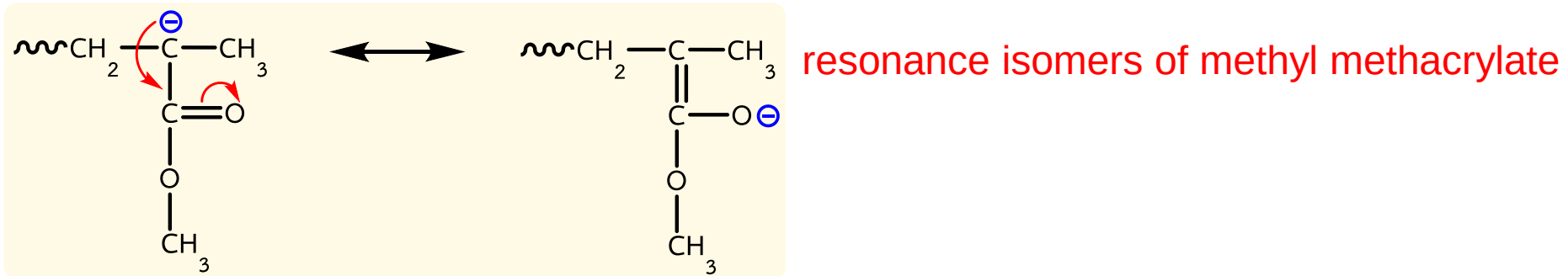
# Anionic Polymerization

# Anionic polymerization

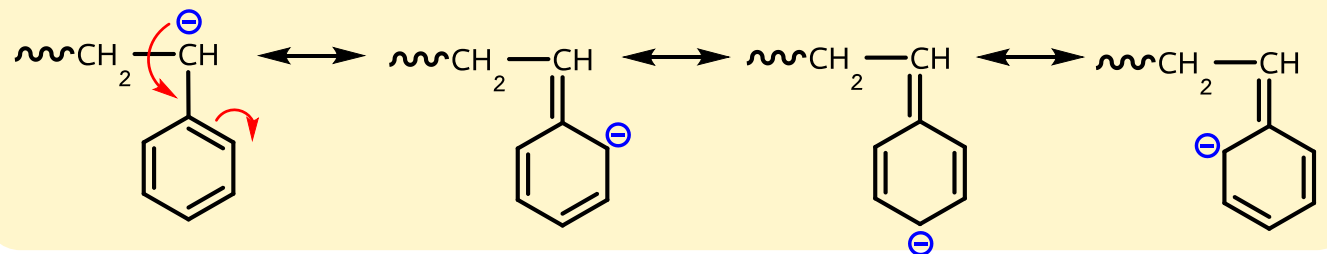


Y = e<sup>-</sup> withdrawing group (-CN, -C(=O)OR, -Ph)

- The initiator is an **anion** (an ion with a negative electrical charge).
- Electron withdrawing groups (ester, cyano) or groups with double bonds (phenyl, vinyl) can stabilize the propagating species by resonance.

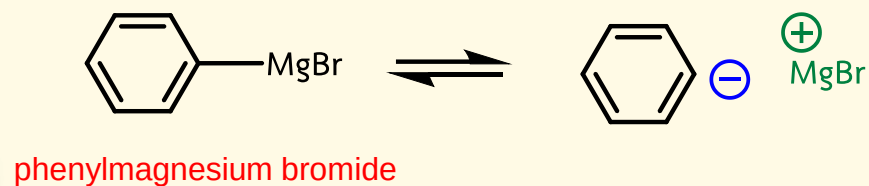
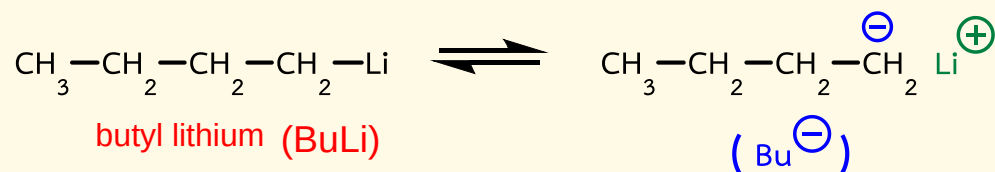


resonance isomers of styrene

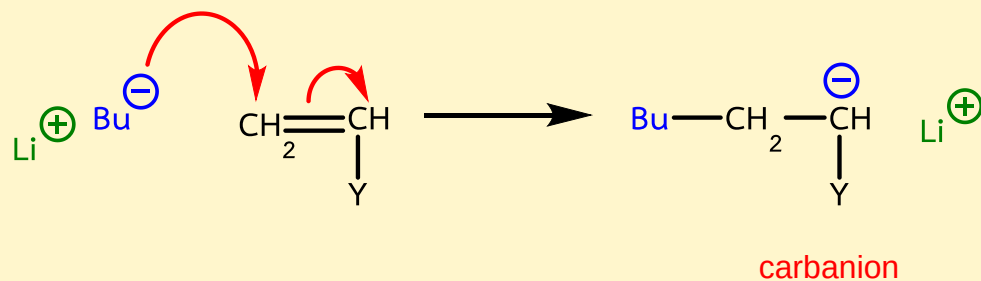


# Steps of reaction

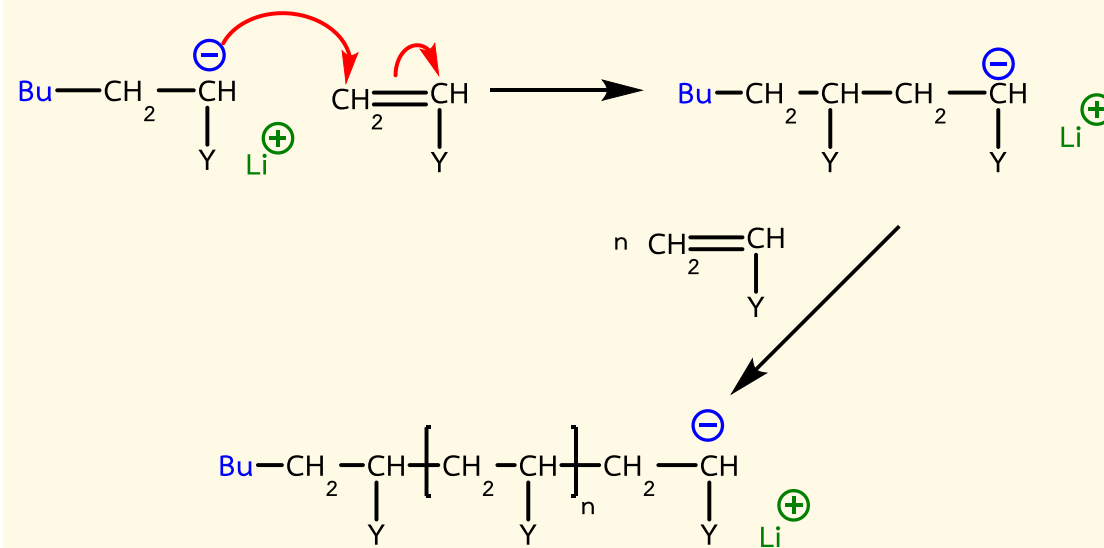
## 1. Initiator-fragmentation step



## 2. Initiation step

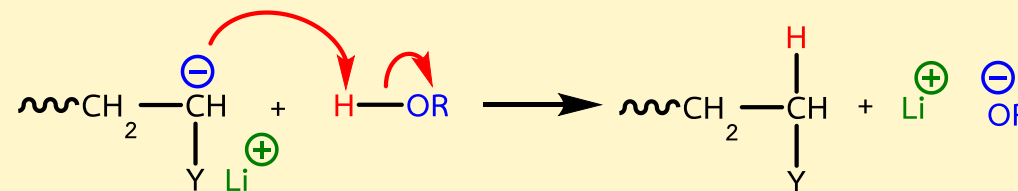


## 3. Propagation step

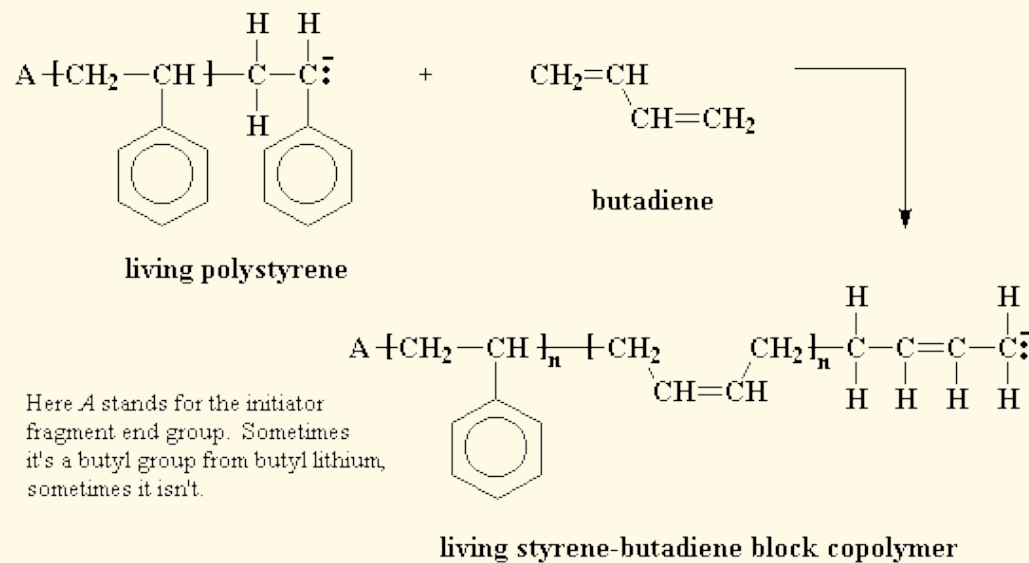


The chain wouldn't die: *living anionic polymerizations*.

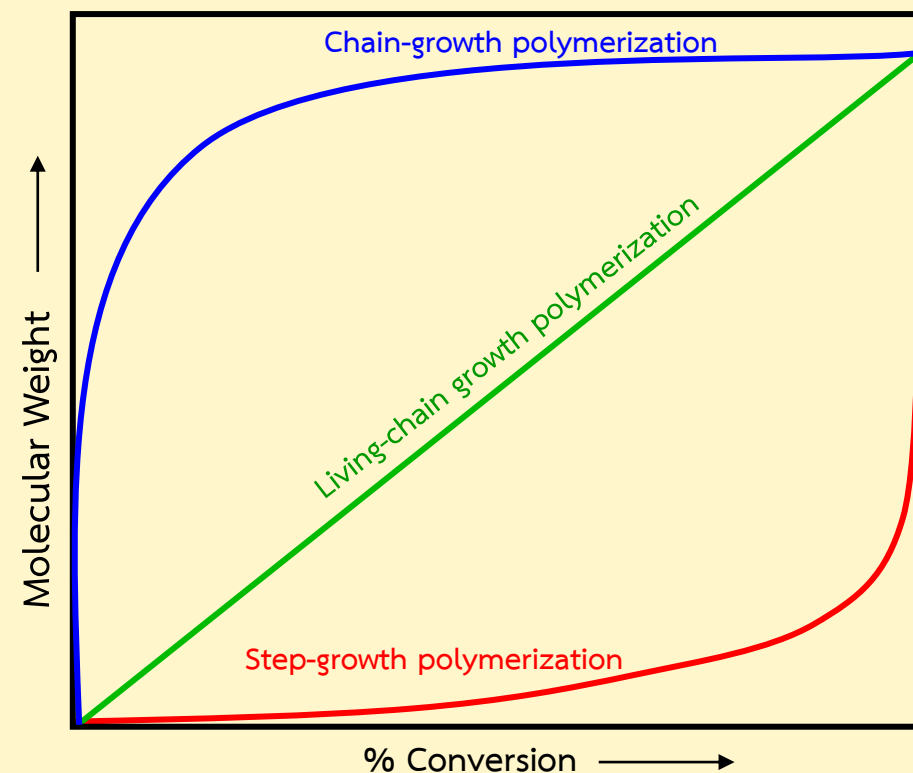
Terminate reaction by impurities (i.e. water).



## 🌸 Synthesis of block copolymer



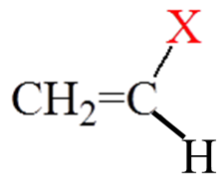
🌸 The plot shows how the molecular weight of polymer changes with the conversion of polymerization reaction.



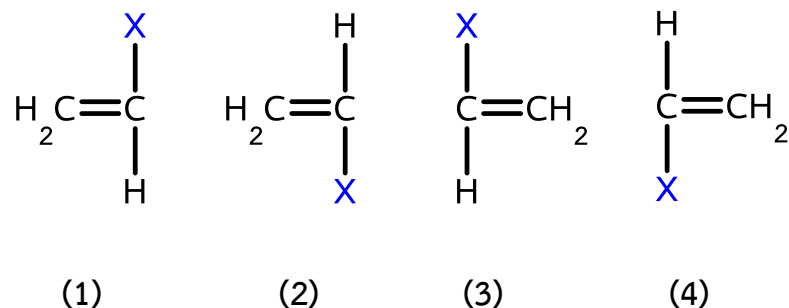


# Coordination Polymerization

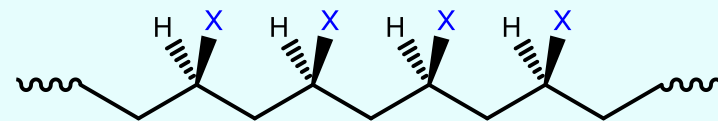
# Tacticity of polymer



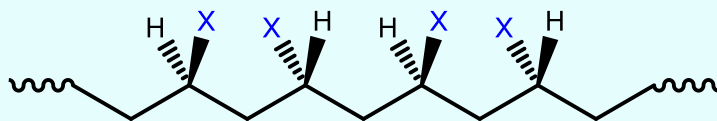
❁ The possible steric configurations of an asymmetric vinyl-polymer chain can be best represented by drawing the chain in its extended-chain or planar zigzag conformation.



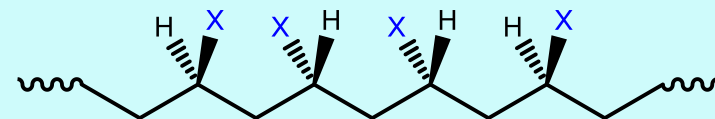
❁ **Isotactic:** All asymmetric carbons have same configuration.



❁ **Syndiotactic:** Asymmetric carbons have alternate configuration.



❁ **Atactic:** Asymmetric carbons have randomly alternate configuration.



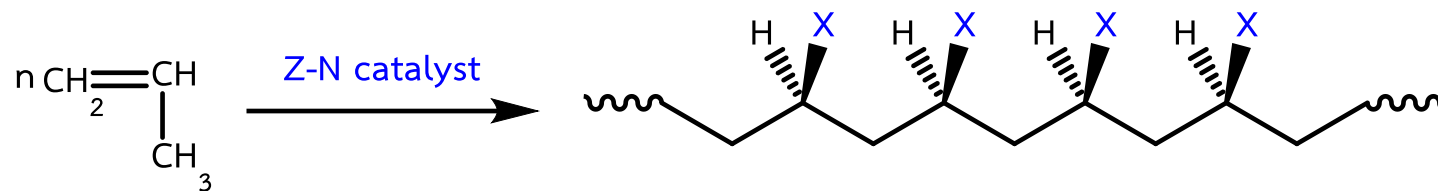
# Effect of tacticity on physical properties of polypropylene

| Property                            | Isotactic | Syndiotactic | Atactic   |
|-------------------------------------|-----------|--------------|-----------|
| Density (g/cm <sup>3</sup> )        | 0.92-0.94 | 0.89-0.91    | 0.85-0.90 |
| Melting temperature (°C)            | 165       | 135          | -         |
| Solubility in hydrocarbons at 20 °C | none      | medium       | high      |
| Yield strength                      | high      | medium       | very low  |

| Polymer | $T_g$ atactic | $T_g$ syndiotactic |
|---------|---------------|--------------------|
| PP      | -20°C         | -8°C               |
| PMMA    | 100°C         | 120°C              |



# Ziegler-Natta catalysts



a transition metal complex with organometallic compound

**Halide compounds of transition metal (catalyst):**  $\text{TiCl}_4$ ,  $\text{TiCl}_3$ ,  $\text{VCl}_4$

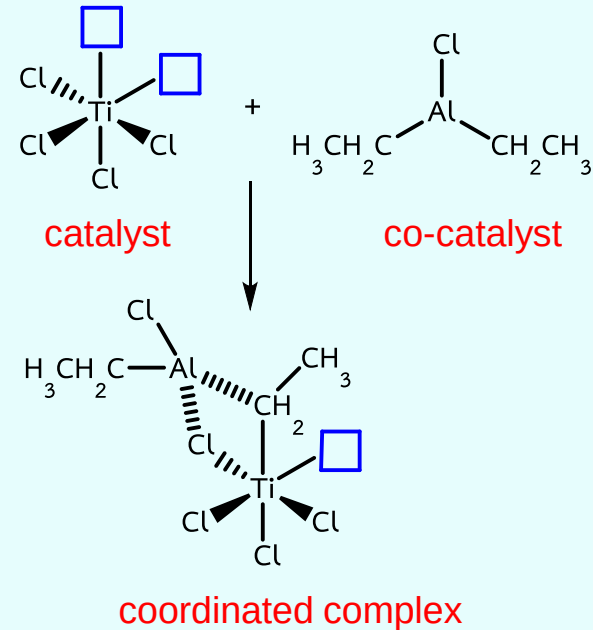
**Organometallic compound (co-catalyst):**  $(\text{C}_2\text{H}_5)_3\text{Al}$ ,  $(\text{C}_2\text{H}_5)_2\text{AlCl}$

- ✿ **Titanium** compounds are the most isoselective.
- ✿ **Vanadium** compounds are the most syndioselective.

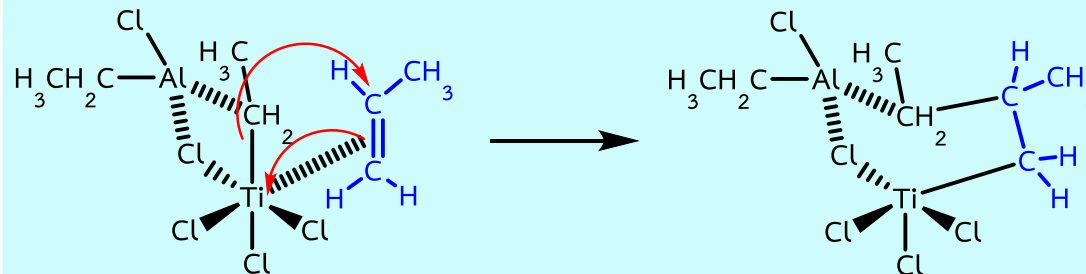


# Isotactic polymerization

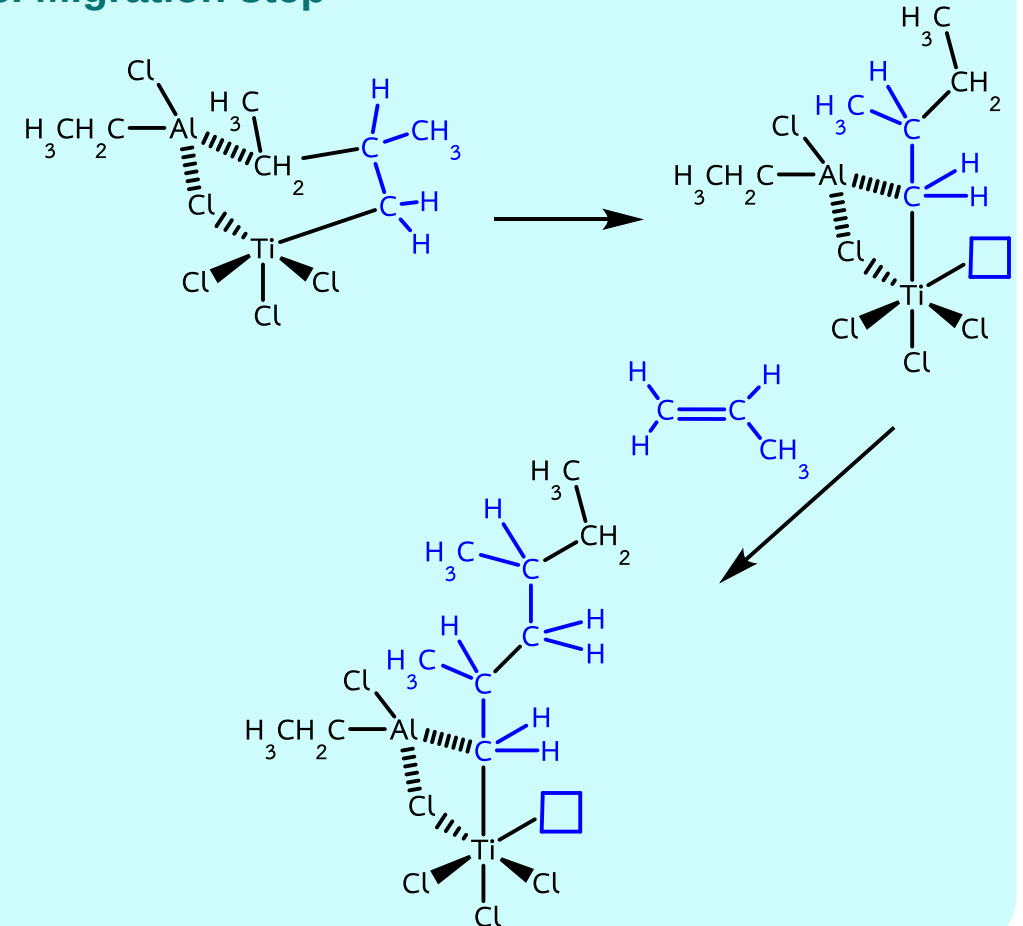
## 1. Initiator formation



## 2. Coordination step



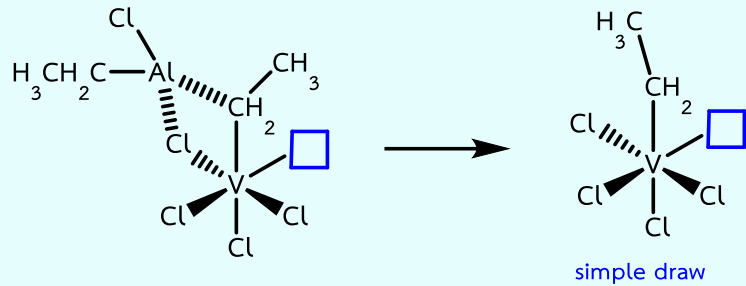
## 3. Migration step



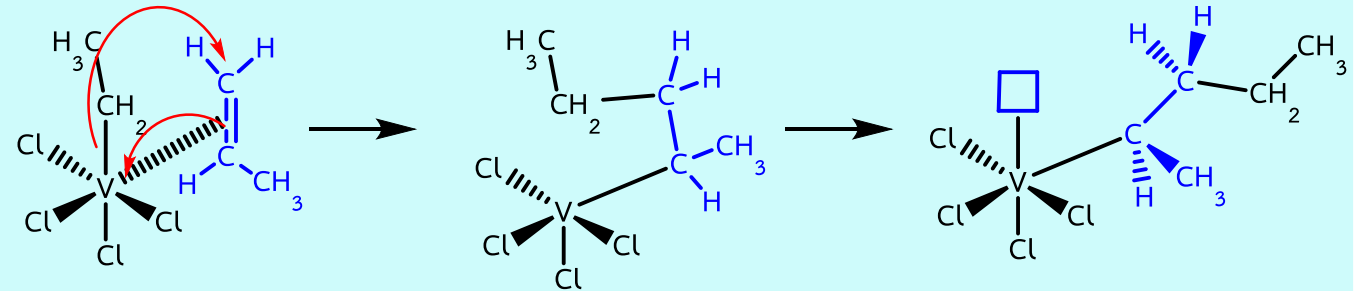
**Catalyst site control model:** All the methyl groups on the growing polymer are on the same side of the chain given isotactic polypropylene.

# Syndiotactic polymerization

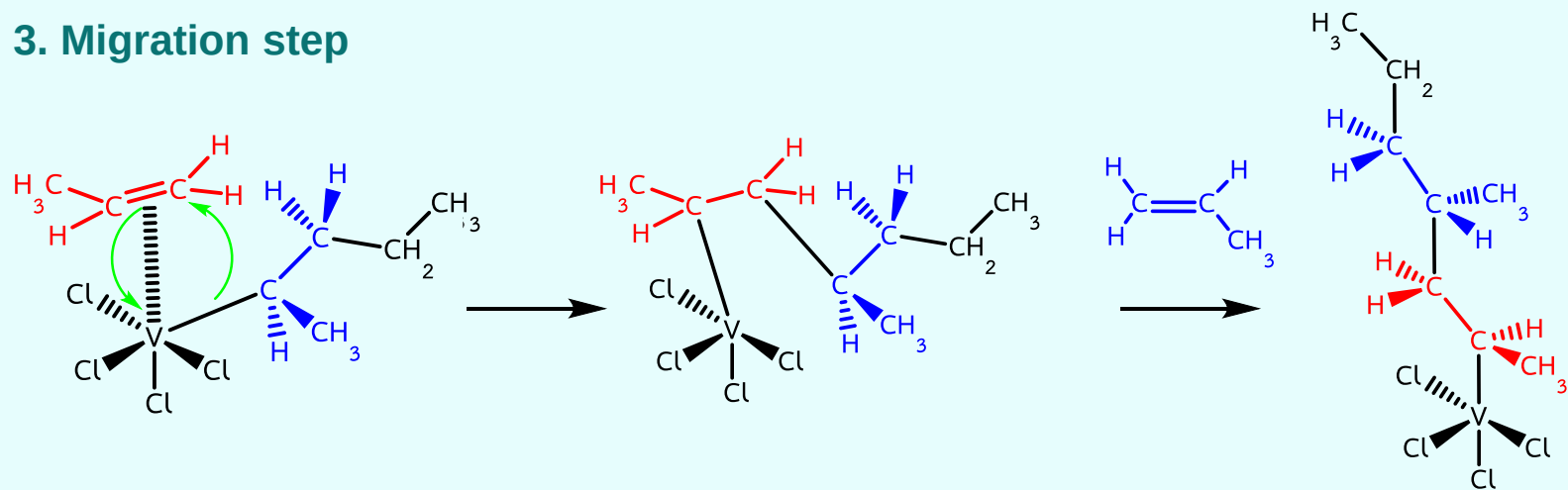
## 1. Initiator formation



## 2. Coordination step



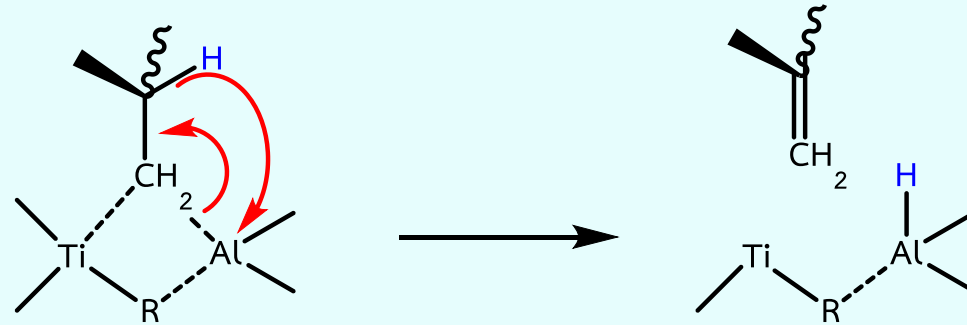
## 3. Migration step



**Polymer chain end control model:** The growing polymer chain stays in its new position until another propylene molecule comes along.

# Termination step

(a)  $\beta$ -hydride shift



(b) Reaction with  $\text{H}_2$ : molecular weight control

